Elemental Battlegrounds implementation specification

Mac Studio hosted multiplayer for iPad browsers

Version 1.0 • Research cut off 1 November 2026 • Implementation contract

Build a private, browser playable, third person 3D free for all for Arel and Nora. Run the authoritative game server on the Mac Studio and render the game locally in each iPad browser. Use floating islands and an original anime inspired visual identity. Preserve the reference game’s recognizable action vocabulary, four slot kits, directional dashes, defensive timing and awakening structure.

This specification defines a complete implementation path, not game code. The initial playable release contains two fully specified fighters; the next milestone adds a third. Across these milestones it specifies 24 slotted abilities, three awakening activations, three contextual attacks, universal combat, and the relevant variants and passives. Every shipped name, character, animation, icon, sound and environment must be independently created.

The reference is Roblox The Strongest Battlegrounds by Yielding Arts. The official experience confirms mobile support and the core control actions. Detailed moves and variants are documented by community sources, with conflicts recorded here. No live Roblox iPad session or frame measurement was performed. This is therefore a researched compatibility target, not a certification of exact November 2026 parity. [R01–R10]

Main implementation decision

Implement the precise local rules in this document. Keep original game observations in a separate reference ledger. Do not fill gaps by inventing facts about the reference. Local tuning values are deliberately specified so the engineering work is not blocked by missing proprietary timings.

Intended implementers

Sonnet 5.5 Low and Sol 6.1 Low are the requested execution targets. Treat these labels as the user’s agent selections, not a claim about model availability. Give each agent one numbered task with its dependencies, files and acceptance checks. The embedded AGENTS.md governs implementation work.

Delivery contents

Product and touch UX • source backed move catalog • executable rules and data contracts • authoritative simulation • project structure • deployment and security • performance budgets • test matrix • sequenced commits • verification ledger • source bibliography.

Contents

Use this map to find the implementation contracts. Headings are also available in Word’s navigation pane. All local gameplay numbers are design values unless explicitly identified as source observations.

1 Scope and document authority
2 Research findings and verification limits
3 Reference catalog for The Strongest Hero
4 Reference catalog for Hero Hunter
5 Reference catalog for Destructive Cyborg
6 Player journeys and screen behavior
7 iPad touch layout
8 Touch gestures and input ownership
9 Camera settings and feedback
10 Floating island arena contract
11 Universal movement and combat values
12 Authoritative state machines
13 Hit resolution and combat invariants
14 Ability data schema and defaults
15 Stone Vanguard base abilities
16 Shared variants passives and handler tests
17 Architecture and dependency baseline
18 Project tree and module ownership
19 Network message contracts
20 Replicated state and HTTP interface
21 Fixed simulation loop and movement
22 Prediction interpolation and reconnection
23 Rendering animation assets and audio
24 Performance budgets and hosting
25 Automated acceptance tests
26 Device network and performance acceptance
27 Sequenced commits for the playable release
28 Agent readable AGENTS instructions
29 Assumptions verification log and deviations
30 Game research sources

1 Scope and document authority

Release priorities

Milestone	Required content	Exit condition
P0 playable release	Two iPads, one private room, floating arena, universal combat, Stone Vanguard and Tide Duelist, all 16 slotted moves and both awakenings	Two people complete the device acceptance suite with no input or authority failures
P1 complete scoped release	Ember Artificer and its eight moves, all three contextual Vanguard attacks, props and contextual wall attacks, final original assets	All P0 and P1 tests pass and asset provenance is complete
P2 separately approved expansion	Other reference fighters, alternate Monster Form, event or restricted kits, ranked modes, additional reported burst systems	New source ledger and complete ability contract before implementation

P0 and P1 together are the full scope of this document. Do not add a new character because a guide mentions it. Arel and Nora can choose the same fighter. No purchases, loot boxes, account registration, public chat, voice, public matchmaking, mobile app wrapper, persistent competitive ranking or Roblox integration are required.

Normative language and evidence

MUST means acceptance requires it. SHOULD is a preferred implementation with a documented equivalent allowed. Every rule without a source tag is a local product decision, marked D by default; it is not a claim about The Strongest Battlegrounds.

Tag	Meaning	Permitted use
V	Verified in an inspected primary source	Cite precisely; do not extend beyond the source
C	Community documented, not directly measured	Use for reference behavior with stated freshness limits
X	Conflicting or incomplete evidence	Keep alternatives in the ledger; do not call one exact
D	Deliberate local design value	Implement as written and expose in configuration
U	Unknown or unverified	Do not ship as a claim of original fidelity

Order of authority

Explicit product requirements → this document’s local rules and numeric registry → ability overrides → default handler semantics → task acceptance tests. A contradiction must produce a short decision record and a corrected test, not silent improvisation. Source descriptions never silently override local configuration.

How to use the document

Read sections 1–6 before scaffolding. Implement sections 7–16 as the gameplay contract, sections 17–24 as engineering and hosting contracts, and sections 25–29 as delivery gates. Source IDs resolve in section 30. The page headings form a navigable outline in Word.

All distances are metres in the new game. Time in simulation tables is integer ticks at 60 Hz. Health is 100 points. Reference percentages are reported separately and are never converted from Roblox studs without a measurement.

2 Research findings and verification limits

What the primary source actually establishes

The official Roblox experience names Yielding Arts, supports mobile, PC and console, and lists punch, block, dash, ragdoll cancel, run, awakening and emote actions. It does not establish iPad coordinates, touch target sizes, hitbox dimensions, damage schedules or frame data. The indexed page showed an update date of 17 October 2026; other aggregators reported later dates. Treat this as an indexing discrepancy, not proof of the live build. [R01]

Research method

Research used the official experience page, community character and combat pages, contemporary control guides, update archives, and primary engine documentation. Search indexed excerpts were used where direct Fandom fetches failed. These excerpts were available on 1 November 2026, but many character pages were last crawled roughly four months earlier. A recently crawled derivative guide does not independently verify a changed move.

Question	Finding	Status and consequence
Action mapping	Core actions confirmed by creator listing; mobile guide maps them to touch controls	V for action set, C for touch mapping [R01 R02]
Exact default iPad layout	No authoritative pixel map obtained; players report customization	U; use measured local layout and calibration procedure
Three core kits	Four base and four awakening slots documented per selected kit	C; scope is explicit [R03 R05 R07]
New contextual attacks	Vanguard reference has Slap, Sneeze and Trashcan entries	C; implement in P1 [R04 R09]
Numeric balance	Hero values, counter rules and cyborg duration differ across pages	X; use D values, preserve conflict log
Current entire roster	Sources count public, alternate and restricted entries differently	U; no claim of complete current roster
Engine APIs	Colyseus documentation now describes version 0.18 APIs	V; use current contract, not old tutorials [T01–T03]

Required fidelity check before calling the game a close reproduction

On each actual iPad, record model, iPadOS, viewport, browser chrome, Roblox build observation date and the original game’s control settings. Capture the default layout and one customized layout with permission. In a private reference session, measure 20 trials per input and 10 per move; record median, range and frame rate. Compare placement, hold behavior, charge release, evasive reuse and fourth punch variants. Store observations in docs/reference-observations.md and normalized rectangles in docs/control-calibration.json. This gate affects fidelity claims; it does not block implementing the D defaults.

3 Reference catalog for The Strongest Hero

The following is a concise record of community reported mechanics, not a balance promise. Names in this section are research labels only. The game’s shipped names appear in the right column. The base and awakening catalog is corroborated across character pages. [R03 R04 R09]

Reference move	Reported behavior	Independent move ID and label
Normal Punch	Heavy direct strike with a ranged shock component	sv.b1 Fault Strike
Consecutive Punches	Repeated forward hits	sv.b2 Gravel Flurry
Shove	Short push; can reflect some projectiles	sv.b3 Rift Palm
Uppercut	Upward launcher	sv.b4 Rising Pillar
Death Counter	Reactive lethal counter	sv.a1 Stillpoint Reversal
Table Flip	Large ground upheaval	sv.a2 Skybreak
Serious Punch	Large directional shock attack	sv.a3 Horizon Break
Omni Directional Punch	Capture followed by attacks from multiple directions	sv.a4 Orbit Collapse

Serious Mode is reported as persisting until an awakening skill is used. Sources describe unusually fast meter gain and no immediate activation heal. One source says twice the meter rate; another reports three times. These are not reconciled as exact values here. [R03 R04 R11]

Additional reported context

The newer character page lists Serious Slap, Serious Sneeze and Serious Trashcan as mutually exclusive contextual uses within an awakening, in addition to a slotted awakening skill. Map these to sv.x1 Echo Tap, sv.x2 Gale Rebuff and sv.x3 Meteor Cast. Their existence is community documented; their exact live triggers require an iPad reference check. [R04 R09]

Numeric conflicts worth preserving

The older page reports Normal Punch as 30 percent total including landing damage, and Consecutive Punches as 15 percent. The newer page reports Consecutive Punches up to 14.5 percent and Shove as 6.9 percent. Do not average contradictory numbers or label the local values as measured. [R03 R04]

Identity brief

Stone Vanguard uses a broad, asymmetrical stone mantle, cropped dark hair, a charcoal and amber palette, bare stone bracers and angular footwork. No bald yellow suited hero, red glove costume, recognizable facial design, signature cinematic composition, copied typography or sampled voice/music. The power fantasy is geological force. Original attack names in this document remain working names subject to a distinctiveness review before public release.

P0 includes all eight slotted moves and the awakening. P1 adds all three contextual moves, a wall interaction and cosmetic knockout finishers. All local timing and damage values below are D.

4 Reference catalog for Hero Hunter

This scoped kit is the standard Hero Hunter, not its separately listed Monster Form. The character page flags unfinished and rework related content, so archived variants must not be mistaken for confirmed live behavior. [R05]

Reference move	Reported behavior	Independent move ID and label
Flowing Water	Capture into a sequence of strikes	td.b1 Current Bind
Lethal Whirlwind Stream	Grab, drag and slam	td.b2 Undertow Spiral
Hunter’s Grasp	Grab and throw	td.b3 Tidal Cast
Prey’s Peril	Counter stance; projectile reflection	td.b4 Mirror Current
Water Stream Cutting Fist	Awakened barrage ending in a slam	td.a1 Cascade Chain
The Final Hunt	Launch and aerial follow up slam	td.a2 Deepfall
Rock Splitting Fist	Powerful chop with a follow up	td.a3 Split Tide
Crushed Rock	Advancing capture and impact, with variants	td.a4 Riptide Pursuit

Rampage replaces the four slots; a community page reports 49 seconds, a damaging activation, armor during successful grabs and a finisher healing passive. These are C observations, not proprietary implementation details. The new game uses different presentation and explicitly defines which attacks grant armor. [R05]

Variants and limits

A contemporary derivative guide describes a close range Crushed Rock assault chain, a ranged variant, and a character specific Martial Artist interaction. It also distinguishes a missed first hit of Rock Splitting Fist and lists removed moves Hunter’s Cruelty and Doomed Prey. P1 includes local close and ranged branches and the missed chop branch. The Martial Artist interaction is excluded because that fighter is outside scope. Removed moves are not implemented. [R06]

The guide reports Flowing Water 24.5 percent, Lethal Whirlwind Stream 18.5 percent, Hunter’s Grasp 8.3 percent with collision dependence, and Crushed Rock 27 percent. These values are reference observations only. Exact finisher thresholds, wall geometry exploits and all reported combo techniques are not a compatibility guarantee. [R06]

Identity brief

Tide Duelist wears layered ivory and teal fabric over dark trousers, with a circular ceramic shoulder guard, tied copper hair and an upright open palm stance. Effects are translucent ribbons and refracted droplets, not a copied martial arts costume, hairstyle, aura, choreography or anime face.

P0 includes all eight slotted moves and awakening. P1 adds the defined variant branches, wall interaction and original knockout effects. The default P0 branches still follow the complete local ability contract.

5 Reference catalog for Destructive Cyborg

This ranged pressure kit is P1. Its inclusion adds charge and release input, directed aerial movement and beam attacks, which must pass the same authority tests as melee. [R07 R08]

Reference move	Reported behavior	Independent move ID and label
Machine Gun Blows	Barrage followed by a launching hit	ea.b1 Cinder Array
Ignition Burst	Propulsion with a finishing blast; early second press	ea.b2 Vent Rush
Blitz Shot	Charged projectile, released by letting go	ea.b3 Ember Orb
Jet Dive	Rise then descending area impact	ea.b4 Furnace Drop
Thunder Kick	Advancing kick and follow up; miss explosion branch	ea.a1 Arc Descent
Speedblitz Dropkick	Fast capture into aerial slam	ea.a2 Comet Lock
Flamewave Cannon	Sustained directional beam	ea.a3 Furnace Ray
Incinerate	Large one use beam	ea.a4 Solar Vent

Maximum Energy Output is a damaging transformation. One community page reports 42 seconds and another 47 seconds. Sources also describe an extended fourth basic attack, a damaging grounded evasive and a finisher cooldown reset. [R07 R08]

Conflicting details

The newer page excludes Flamewave Cannon and Incinerate from the reset passive. Counterability of the grounded evasive differs between pages. Thunder Kick is listed as 40 or 41 percent; Speedblitz Dropkick as 50 or 49 percent against standing targets. The local configuration intentionally resolves these rather than claiming live numerical parity. [R07 R08]

Variants included in the local contract

Ember Orb supports partial and full charge; Vent Rush supports airborne aim and an early second press; Furnace Drop supports landing impact and an interrupted fall. Arc Descent has hit, miss and interrupted miss branches. Comet Lock has a lower damage branch for ragdolled targets. Furnace Ray and Solar Vent specify sweep limits and repeated hit caps. The cyborg and hero joint awakening cinematic is excluded: it is a presentation interaction, not required to reproduce the combat roles. [R08]

Identity brief

Ember Artificer is a human engineer with a segmented ceramic backpack, floating copper rings and heat vents in independent gauntlets. Use plum, copper and pale orange; no copied cybernetic arms, hair, face, costume, voice or transformation. Beam silhouettes, lens effects and sound envelopes must be original.

Unverified roster and update claims, including new global burst meters or character exceptions, are tracked as P2 research. Do not create generic damage multipliers based on unsourced guide statements.

6 Player journeys and screen behavior

First visit and joining

1. Boot loads a small shell, checks WebGL2 and WebAssembly, and displays “Preparing arena” with byte based asset progress. Unsupported hardware shows a readable explanation and Retry, with no join request.

2. Welcome shows Play, Practice, Settings and a connection status. Arel and Nora are local preset labels; selecting a name never authenticates a player. An adult supplied invitation authenticates the session separately.

3. Play exchanges the invitation for a session, then shows a private lobby with room status, fighter selection and Ready. Default room capacity is 4; target use is 2. Seats include reconnect reservations.

4. Selecting a fighter shows its four base moves, awakening overview and a 3D original model. Unimplemented P1 kits are hidden in P0, not clickable placeholders.

5. Ready submits the selected kit and content hash. Server validates both, spawns the player and sends the full state. Client enables gameplay only after state plus required assets are ready.

During a session

FFA has no teams and no timed round. A session scoreboard tracks knockouts, deaths and assists. A KO awards one point; falling off an island awards the most recent hostile attacker within 8 seconds, otherwise no scorer. A same tick tie uses the damage resolver order. A player who dealt at least 20 damage in the previous 8 seconds receives an assist if another player receives the KO. Each victim life can award only one KO.

Death shows the scorer and a 3 second respawn countdown. The server chooses a safe spawn and restores health. Kit changes requested in combat take effect at next respawn; a living out of combat player may change after 8 seconds without dealing or taking damage, resetting meter and cooldowns. Never use kit switching to heal or clear a combat timer.

Menus and interruption

Settings and the scoreboard do not pause the world. Opening Settings releases every held input and explains “The arena continues while this menu is open.” Leave returns to Welcome and closes the room connection. Reconnect preserves the existing fighter and life state; it does not grant a fresh spawn or invulnerability.

Practice runs a separate server room with one stationary dummy and optional guard/ragdoll toggles. Practice damage and meters never affect FFA. A short tutorial asks the player to move, turn, hold attack, hold block, dash, use all four skills and awaken; each completion is server confirmed.

Error states

ROOM_FULL, INVITE_INVALID, VERSION_MISMATCH, SERVER_OFFLINE and ASSET_FAILED each have plain language text and Retry or Back. Retry uses bounded exponential delay; never loop invitations or silently enter another room. Do not expose stack traces to children.

7 iPad touch layout

All dimensions here are D, not measured Roblox coordinates. The layout preserves the researched control vocabulary and thumb zones. Exact visual similarity is subject to calibration; use original icons and styling. [R01 R02 R10]

Define the safe rectangle from visualViewport and CSS env safe area insets. W and H are its CSS pixel width and height; its origin is x0,y0. Use landscape. Scale s = clamp(min(W/1024,H/768),0.78,1.25). All positions below are centres in the safe rectangle. Final hit targets are at least 48 CSS pixels, separated by at least 8 pixels.

Control	Centre formula in CSS pixels	Visual size
Movement joystick	(104s, H−112s)	Base diameter 136s; knob 58s
Attack	(W−92s, H−190s)	Diameter 76s
Jump	(W−76s, H−80s)	Diameter 68s
Dash or Evasive	(W−180s, H−110s)	Diameter 64s
Block	(W−182s, H−205s)	Diameter 64s
Camera facing lock	(W−78s, H−286s)	Diameter 48s
Skill slots 1 to 4	(W/2 + (i−2.5)90s, H−64s)	82s by 60s
Awakening bar and action	(W/2, H−126s)	320s by 44s
Menu and fighter	(32s,32s), (92s,32s)	48s each
Health and name	(W/2,32s)	240s by 32s
Scoreboard	(W−36s,32s)	48s

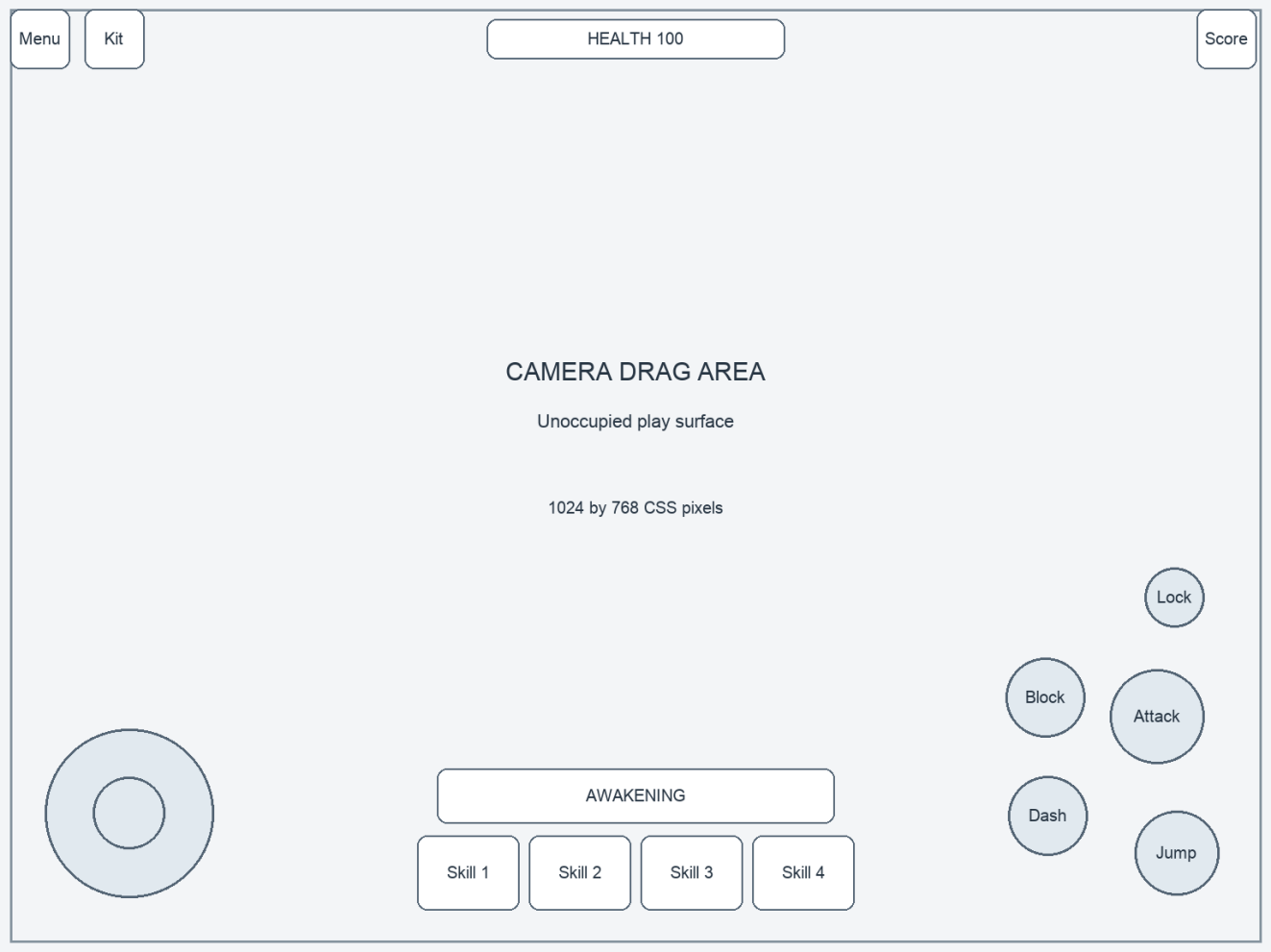
At W=1024,H=768,s=1, attack is (932,578), jump (948,688), dash (844,658), block (842,563), skills (377,704), (467,704), (557,704), (647,704). These coordinates are reproducible acceptance fixtures.

Responsive behavior

For W<900, use s from the formula and reduce the skill strip gap to 6s; rerun the rectangle collision test. If safe width is below 700 or H below 420, show a rotate or enlarge window panel and release inputs. Portrait is supported for lobby and settings; active combat shows “Turn your iPad sideways” and does not pretend that orientation can be forced. The avatar remains vulnerable until normal server timeout rules apply.

Layout editor in Settings allows drag, size 0.85–1.30 times default, opacity 0.35–0.90, left hand mirror and Reset. Reject overlaps and unsafe positions. Persist normalized positions per viewport class and local profile. On rotation, validate and fall back to defaults if necessary.

7 Touch layout reference drawing



The drawing shows the specified local landscape layout at a 1024 by 768 CSS pixel safe rectangle. It is an engineering wireframe, not a screenshot of Roblox and not a verified original default. Use section 7 formulas as the source of dimensions.

Buttons occupy the lower corners and bottom centre. The camera drag region is the unoccupied play surface; it does not overlap a control owner. The joystick stays anchored while the thumb moves. Skill slots use original icons and visible numbers, while awakening has a larger separate action target.

The player may customize this arrangement after calibration. Save separate normalized layouts for wide landscape and compact landscape, retaining minimum target dimensions. Verify the actual iPad viewport after browser bars and safe areas are applied; do not use hardware screen pixels as CSS coordinates.

8 Touch gestures and input ownership

Use Pointer Events with touch action disabled only on the combat surface. Capture each pointer on its initial owner. Release on pointerup, pointercancel, lostpointercapture, blur, visibility loss, orientation change and menu opening. Each pointer owns exactly one control until release; moving over another button does not activate it. [T06]

Input	Local behavior and server intent
Joystick	Fixed centre; 10s pixel dead zone; radial magnitude remapped to 0–1; cap at 56s radius. Send a camera relative horizontal movement vector
Camera drag	Any unowned pointer starting outside controls rotates view. Sensitivity 0.003 rad per CSS pixel divided by s; pitch clamp −55 to +25 degrees
Attack	pointerdown requests M1; holding sets attackHeld. Server repeats at legal combo times. No synthetic browser click action
Block	Hold sets blockHeld; release clears it immediately. No toggle by default. Cannot attack and guard simultaneously
Jump	pointerdown sends edge; held state selects grounded fourth hit uppercut. Holding does not repeatedly jump
Dash	pointerdown sends one edge, with current move vector. While ragdolled it requests evasive. Stationary request defaults forward
Skills	Tap fires on pointerdown in Quick Use mode. Charge skill begins on down and releases on up. A second distinct press triggers a permitted recast
Awakening	One tap requests activation only if the server meter is full; never on pointerup after a cancelled touch
Facing lock	Tap toggles camera aligned facing; client shows a small original reticle. It is not automatic target lock

Multi touch and conflicts

Support at least five simultaneous pointers: move, camera, attack, jump and one skill or dash. UI precedence is button → joystick → camera. The first pointer wins if two pointers claim the same hold control. Ignored extra pointers cannot release the owner’s action. Releasing the camera must not stop movement or guard.

Jump versus fourth M1: on attack commit, airborne selects downslam; otherwise jumpHeld selects mini uppercut; otherwise the normal finisher. During an active combo, holding Jump suppresses the ordinary grounded jump until the combo ends, allowing the uppercut gesture. A jump edge before starting the combo is an ordinary jump. These precise priorities are local decisions.

Optional selection mode

Quick Use is on by default. If off, tapping a skill selects it and Attack activates it once; a charge skill charges while Attack is held. Cancel selection with a second tap of the same slot, guard, another slot or 3 seconds inactivity. Selected mode is local only and does not bypass server checks. Suppress normal attackHeld until that activation pointer is released. The original’s exact settings defaults remain U; an update archive reports held skills with auto use. [R10]

Send neutral intent immediately when input is cancelled. Server additionally clears held actions after 15 ticks without a fresh input packet. Never rely on a pointerup packet reaching the server.

9 Camera settings and feedback

Camera contract

Third person camera pivot is character centre plus 1.1 metres. Default distance 6.5 m, allowed 3.5–10 m via Settings slider; vertical field of view 60 degrees, near 0.1, far 180. Camera sphere cast radius 0.18 m against static geometry pulls inward to the nearest obstruction, with 0.1 m clearance. Ease outward over 0.15 seconds; pull inward immediately. Camera position and shake never affect hit detection.

Facing lock on: horizontal camera yaw determines requested facing, validated to at most 720 degrees/second when not locked by an ability. Facing lock off: moving turns the avatar toward movement at 540 degrees/second; while stationary keep last facing. Ability commit uses authoritative facing, except specifically allowed aim updates. No invisible aim assist, enemy magnetism or automatic camera target selection.

A cinematic is a local presentation of a server event. Captured participants may receive a short original camera track; other players keep normal camera control. Reduced Motion uses the ordinary camera. Cancel the track immediately on victim death, respawn, disconnect, menu, or scene disposal. The world never pauses for a cinematic.

Feedback timing

Button highlight begins on local input within one render frame. Cooldown tint may show a pending outline, but the countdown begins at the server accepted tick. Rejection clears the pending outline and shows one of “Recovering”, “Cooling down”, “Need full meter” or “Cannot use now” for 0.7 seconds. Reject messages are rate limited to one per reason per second.

Confirmed hits create distinct light, heavy, blocked and countered sound/visual cues. Predicted attacks can animate a windup but cannot show confirmed damage, hitstun, KO or meter gain. Damage numbers default off. The awakening bar shows integer percent and “Ready”; all cooldowns use one decimal under 10 seconds and ceiling integer seconds above it.

Accessibility and comfort

Default camera shake is 0.25 of authored amplitude, flash intensity 0.3 and motion blur off. Reduced Motion disables shake, screen warps and forced cameras. Color is reinforced with shape and text. Combat hit targets are 48 px minimum; menu targets are 44 px minimum. Sound sliders are independent for effects and music, both bounded 0–1. Audio starts only after Play or another user gesture and resumes after returning from background. No reliance on browser vibration.

Settings persist locally with a schema version and validated ranges: layout, sensitivity, facing lock, Quick Use, auto run, handedness, volume, quality, reduced motion and flash level. Export/import settings JSON is capped at 8 KB and accepts only known keys. Do not store invitations in these settings.

10 Floating island arena contract

Geometry and units

Use Y up, metres, and local forward −Z. Main island walkable top is a 48 by 40 m rounded rectangle at y=0, thickness 5 m. Four satellite islands are 12 by 12 m at centres (−36,0,0), (36,0,0), (0,0,−32), (0,0,32). Connect with 4 m wide bridges: 6 m east and west, 6 m north and south. Bridges are permanent static colliders. Guardrails appear only at spawn decks; combat edges remain exposed.

Use simple invisible collision hulls distinct from decorative meshes. Static obstacles on the main island: four 3 by 2 by 2 m pillars at (±12,1,±10), two waist high walls at (−10,0.6,0) and (10,0.6,0), each 4 by 1.2 by 0.6 m. Keep the central 12 m diameter area clear. Satellite islands contain no hard cover in P0.

Spawns and falls

Eight spawn anchors are (±18,0,±14) and (±34,0,±3). Spawn capsule centre is surface height plus 0.9 m. Reject anchors within 2 m of occupied capsules or within an active hostile hit region; choose the remaining anchor maximizing distance to the nearest living enemy. Ties use lowest anchor ID. If none qualify, retry for 60 ticks, then choose the farthest and grant the standard spawn protection.

Void plane y=−25 causes immediate KO once per life; radial horizontal boundary 90 m does the same. Falling itself deals no generic damage. The specific Fault Strike landing bonus is separate. Knockback can cause ring outs. Jump and mobility abilities cannot create a safe invisible floor. A grab must abort if its next position intersects a solid wall; release the victim at the last valid point.

Destruction and props

P0 destruction is cosmetic. No ability removes bridge or island collision. P1 adds eight server owned throwable stone urns at centres (±9,0.25,±10) and (±15,0.25,±10), each with radius 0.25, mass presentation only, and 10 second respawn. One can be picked up using the contextual Attack rule. Thrown urns use authoritative swept projectile queries, not unconstrained debris physics.

Maximum 80 cosmetic fragments per client, lifetime 2 seconds, no gameplay collision. Pool shards and dust. Large ground attacks deform a visual decal or temporary mesh layer only; they never invalidate navigation or spawn anchors.

Map artifact acceptance

assets/maps/islands.v1.json stores collider primitives, spawn anchors, urn anchors and content hash. Client and server load the identical collision description. A build check verifies bridge connectivity, capsule headroom, spawn containment and all anchors above the void. Keep silhouettes and architecture independent from the reference map.

11 Universal movement and combat values

The original held attack and fourth hit vocabulary is community documented. [R12] Every number on this page is D. Values are tuned for readable tablet play, not claimed as original frame data. Keep them in packages/shared/src/config/combat.ts.

System	Exact local rule
Health and movement	HP 100; walk 5 m/s; run 8; guard 2; acceleration 45 m/s²; air control 30 percent; gravity −24 m/s²
Capsule and jump	Radius 0.35 m, cylindrical half height 0.55; total height 1.8 m; jump speed 8 m/s; no double jump; 6 tick coyote and jump buffer
Ground control	Offset 0.02 m; step height 0.3; minimum step width 0.2; max slope 45 degrees; snap down 0.2; no snap while rising
Basic chain	Four hits: 3,3,4,4 HP. Each has startup 8, active 3, recovery 10 ticks; fourth recovery 30. Gap >54 ticks resets chain
M1 reach	Box width 1.5, height 1.7, depth 1.8 m; front face at 2.1 m from centre; one hit per target per swing
Hit effects	First three M1 hits stun 24 ticks; fourth ragdolls 72 ticks and adds velocity (forward 8, up 3) m/s
Fourth variations	Mini uppercut: 4 HP, up 10 m/s, ragdoll 72. Air downslam: 4 HP, down 16 m/s, radial landing radius 1.5 m, no second damage
Forward dash	18 ticks, speed 22 m/s, last 3 ticks have M1 sized hitbox; 4 HP, stun 24; cooldown 300 ticks
Side dash	12 ticks at 24 m/s; no attack; shared left/right cooldown 120 ticks; no invulnerability
Back dash	18 ticks at 18 m/s; no attack; cooldown 240 ticks; no invulnerability
Evasive	Legal after 12 ragdoll ticks; 12 ticks at 20 m/s; invulnerable first 8; cooldown 1800 ticks
Guard	Frontal 120 degree cone; zero default chip; blocks only attacks marked B; startup 3 ticks; recovery 3 ticks

Auto run defaults on for joystick magnitude ≥0.9 for 6 ticks; stops below 0.75. With auto run off, a second forward joystick excursion above 0.9 within 18 ticks requests run; release below 0.2 cancels it. Dash type uses the larger absolute forward/right component of the joystick relative to authoritative facing; ties choose forward/back. Keep normalized diagonal direction. No input means forward; stationary evasive uses facing right. Keyboard fallback: WASD, mouse attack, F guard, Q dash, Space jump, 1–4 abilities, G awakening, Shift facing lock, Esc menu. Double W requests run.

No passive health regeneration in P0/P1. Spawn protection lasts 120 ticks, cancels on any offensive request accepted, and prevents dealing or receiving damage. Death resets meter, held inputs, capture, combo and all cooldowns. Basic M1, dash and abilities share recovery permission; no animation cancellation unless a handler explicitly allows it.

12 Authoritative state machines

Use orthogonal fields rather than a single enum for every combination. locomotion is Grounded or Airborne; action is Idle, Guard, Dash, Windup, Active, Recovery, Stunned, Ragdolled, Captured, Awakening or Dead; form is Base or Awakened; connection is Active or Dropped. An action instance contains startTick, endTick, moveId, phase, targetId and generation. All deadlines are exclusive: a state is active while tick < endTick.

From	Trigger and condition	Transition and side effects
Idle	Legal ability intent; cooldown ready	Windup; commit charge/cooldown; assign cast ID; snapshot aim
Windup	tick reaches activeStart	Active; create scheduled hit phases
Active	Last active tick completes	Recovery; remove hit volumes and release capture if specified
Recovery	tick reaches endTick	Idle; consume one still valid buffered intent
Idle	blockHeld and grounded	Guard after 3 startup ticks; disallow attack until guard release recovery
Idle	Dash request and legal cooldown	Dash; consume corresponding cooldown; airborne horizontal dash is allowed
Any vulnerable action	Valid hit with stun	Stunned or Ragdolled; cancel current cast unless protected
Stunned	Timer expires	Idle; ground or air state continues independently
Ragdolled	Valid evasive after minimum	Dash with evasive flag; remove pending landing bonus
Ragdolled	Timer expires and grounded	Recovery 12 ticks then Idle; airborne remains ragdolled
Any capturable state	Capture wins arbitration	Captured; suppress locomotion and all intents except disconnect
Idle	Full meter awakening request	Awakening; form switches at completion; no attack during activation
Any living state	HP reaches zero or void rule	Dead; cancel all casts, captures and projectile ownership effects
Dead	Respawn deadline and valid anchor	Idle, Grounded, Base; lifeId increments; apply protection

Interrupt ordering

At the beginning of a tick apply deaths from the prior tick and administrative removals, then expiries, input intents, locomotion, collision queries, hit resolution and newly caused deaths. Within hit resolution, invulnerability wins; then a valid counter; then guard; then damage and reaction. Armor ignores stun/ragdoll from attacks without armorBreak, but still takes damage. armorBreak bypasses armor, not invulnerability. Captured fighters can take third party damage only where the capture handler permits it.

Input priority within one player and tick: evasive → awakening → skill lowest slot index → dash → M1 → jump → guard → movement. An illegal higher priority request does not suppress a legal lower request. Buffer one offensive edge for 6 ticks; replace with the most recent edge, tie by lowest action number. Never buffer awakening, dash, recast or counter stance. Held M1 is not an unbounded queue.

Action cancellation invalidates scheduled effects with the cast generation token. No JavaScript timeout controls damage. A victim death or departure releases its captor immediately into 18 ticks recovery; no stale joint or permanent lock may remain.

13 Hit resolution and combat invariants

Geometry and ordering

Evaluate server poses at the current simulation tick. P0/P1 do not rewind hitboxes; client supplied timestamps cannot make historical attacks. Broad phase uses Rapier queries; exact acceptance uses attack shape intersection, target state and static occlusion. Sweep moving attacks from their previous to current position so a dash cannot tunnel through a target. Exclude self and the same life's already hit entries.

Collect candidate hits, then sort by time of impact within the tick, attacker entity ID, cast counter and hit index. Apply that order serially. Later attacks from a player killed or interrupted earlier in the same tick are invalid. This is a deliberate deterministic tie rule; simultaneous trades are not promised.

Guard and counter rules

For guard, flatten the vector from target to impact origin, normalize, and compare dot(target.forward, towardImpact) ≥ cos(60°). The attacker's screen position is irrelevant. A zero length vector counts as frontal. B means blockable with chip 0 unless overridden; U means bypass guard. Neither implies armor break. A blocked hit consumes its per target hit entry and adds 6 ticks blockstun. It never builds meter.

A counter can intercept only a hit marked counterable and arriving during its open interval. It consumes the incoming hit. Reflection changes projectile owner and direction toward the original owner's current position, preserves damage, assigns a new cast ID and limits reflection count to one. Reflection never teleports a projectile through a wall. A second reflection attempt absorbs it with no damage. A counter cannot recursively trigger another counter from its response unless explicitly stated; all local counter responses are noncounterable.

Capture arbitration and damage

Only one capture owner per victim. At capture commit test both participants alive, within reach, unobstructed and eligible. First valid sorted candidate reserves the victim. Default capture lasts through the scripted damage schedule, releases on final hit, and makes the victim immune to third party capture but not ordinary third party damage. A handler explicitly marked cinematic grants both participants invulnerability to third parties from capture until release; scheduled owner damage bypasses only that capture protection.

Damage is integer hundredths of HP internally. Clamp damage to remaining HP. Meter gain uses actual HP removed, not overkill, blocked damage, practice damage or self damage. No meter gain while Awakened or during an awakening activation. Store damage attribution per victim life; trim entries older than 480 ticks.

Invariants

HP stays in [0,10000]; one KO per life; one passive reward per victim life; all cooldowns are server ticks; no ability hits the same target more than its declared cap; no nonfinite vectors; no movement across static collision; no orphan captures; no attacks during spawn protection; no hitbox derives from animation bone timing.

Wall attacks require an enemy within 0.6 m of a vertical static collider, normal within 15 degrees of horizontal, attacker within 2 m and a fourth M1 request. Reserve wall attack before normal fourth hit. P0 disables this branch; P1 enables the shared wall script in section 16.

14 Ability data schema and defaults

The following schema is an implementation contract. Generate runtime validation from it. Unknown properties fail validation in server configuration. Reference evidence lives in a separate ledger, never in shipped player facing labels.

```
type MoveId = `${"sv"|"td"|"ea"}.${string}`;
type Phase = "Base" | "Awakened";
type Shape = { kind:"box"; w:number; h:number; d:number }
  | { kind:"sphere"; r:number }
  | { kind:"beam"; length:number; radius:number };
interface AbilityDef {
  id: MoveId; slot: 1|2|3|4|null; form: Phase;
  label: string; handler: string; startup: number;
  active: number; recovery: number; cooldown: number;
  hitOffsets: number[]; damage100: number[];
  shape: Shape; range: number; block: "B"|"U";
  counterable: boolean; armorBreak: boolean;
  targetStates: ("standing"|"air"|"ragdoll")[];
  stunTicks: number; ragdollTicks: number;
  launch: { forward:number; up:number };
  armor: "none"|"startup"|"onHit";
  input: "press"|"holdRelease"|"recast"|"context";
  moveScale: number; turnRate: number;
  oncePerForm: boolean; consumesForm: boolean;
  maxTargets: number; maxHitsPerTarget: number;
  animationId: string; vfxId: string; sfxId: string;
}
```

Defaults applied before the per move overrides

slot follows ID suffix; base prefix b and awakened prefix a. startup/active/recovery are ticks; cooldown starts on accepted initial press, including a miss or interruption. activeStart = startTick + startup; hitOffsets are relative to activeStart. Default one hit at offset 0, full listed damage; maxHitsPerTarget equals the explicit hitOffsets length where a multi hit schedule is given, otherwise 1; active is a query window, not a repeated damage interval. Deduplicate one hit across the window.

Default shape box 2 m wide, 2 m high, depth = range, centred range/2 forward and 0.2 m up. Default targets standing and air, maxTargets=1, stun=24, ragdoll=0, launch=(0,0), armor=none, counterable=true, armorBreak=false, moveScale=0, turnRate=0 after commit. Before commit facing uses section 9. Attacks may activate in air unless a card says grounded. “R72 F10 U4” overrides reaction to ragdoll 72 ticks and velocity forward 10, up 4. Effects use IDs equal to move ID plus .anim, .vfx or .sfx.

Once per form means used bit resets only on a new awakening. consumesForm marks the awakening spent at commit; resolve the current ability, then return Base even if interrupted. A one use miss is still spent. Every table number is D. Handler names below are internal interfaces, not library APIs.

15 Stone Vanguard base abilities

ID and label	Startup active recovery	Cooldown	Damage HP	Range and flags
sv.b1 Fault Strike	24 4 24	1200	20 plus 10 landing	3 m U; box wave 8 m B
sv.b2 Gravel Flurry	10 40 18	900	15 total	2.5 m B; max 3 targets
sv.b3 Rift Palm	8 3 14	600	7	2 m U
sv.b4 Rising Pillar	16 4 20	900	15	2 m U

Fault Strike handler faultStrike

Accept grounded or air; target states include ragdoll for direct strike. Direct hit R72 F14 U4. On direct damage attach landingBonus=10 to that victim and cast, expiring after 120 ticks. Apply it on the first new ground contact after at least one airborne tick; evasive, death or protection clears it. At activeStart also query a forward box width 2, height 2, depth 8 for wave targets not directly struck; wave damage 10, R36 F6 U1, no ragdoll targets. Maximum 4 targets across direct and wave. This is a mutually exclusive damage branch, not 20+10+10 on the same target. Acceptance: direct target cancels before landing and loses exactly 20 HP.

Gravel Flurry handler barrage

hitOffsets=[0,8,16,24,32,39], damage=[2,2,2,2,2,5]. Query each tick group once per target. Intermediate hits stun 12; final hit R60 F7 U2. Supports ragdolled targets; moveScale=0.15, turnRate=90 degrees/second. No capture and no armor; a third party hit may interrupt remaining hits. Maximum damage per target is 15. Acceptance: a target entering only at offset 39 receives 5 HP damage, not the missed hits.

Rift Palm handler reflectPalm

On active ticks, first check reflectable projectile overlap with a 1.8 m radius frontal hemisphere, then melee. At most one reflected projectile per cast; a melee hit is still permitted on one other target. Melee supports ragdoll; stun=30; launch F2 U0, no ragdoll. A reflect does not reset cooldown or grant meter until the reflected projectile actually damages someone. Acceptance: an incoming Ember Orb returns with its existing charge damage and cannot bounce indefinitely.

Rising Pillar handler launcher

Supports standing, airborne and ragdolled targets. On hit R84 F1 U11; no extra damage when landing. A target already captured by another attacker is ineligible. Aerial use applies the same shape relative to the authoritative capsule, not to the camera pitch. Acceptance: changing animation playback speed does not change the active tick or launch velocity.

Base abilities do not reset M1 chain progression. A chain gap greater than 54 ticks still resets normally. Finisher art is triggered only after server confirmed lethal damage and cannot add damage or extend capture.

15 Stone Vanguard awakened abilities

ID and label	Startup active recovery	Use limit	Damage HP	Shape and flags
sv.a1 Stillpoint Reversal	0 600 18	Consumes form	Lethal on response	Counter trigger; U; noncounterable response
sv.a2 Skybreak	90 6 45	Consumes form	100	Box 20 by 8 by 24 m; U; armorBreak
sv.a3 Horizon Break	72 6 42	Consumes form	100 near, 60 far	Beam 32 m radius 3; U; armorBreak
sv.a4 Orbit Collapse	24 6 36	Consumes form	100	Capture range 3 m; U; armorBreak

Stillpoint Reversal handler lethalCounter

Commit marks the form spent and returns the base slot set immediately while leaving a counter status open for 600 ticks. Allow base actions during that status. A valid counterable incoming hit triggers a cinematic capture if the attacker is within 40 m and reachable without a solid wall between the pair; otherwise absorb that hit and end the counter. On trigger, cancel both actions, grant capture protection, and schedule 100 damage at trigger+60 ticks, then release at +72. No trigger means no damage or heal. On successful damage heal caster to 100 HP. A noncounterable hit damages normally and may kill the caster, ending the status. No hidden teleport through terrain.

Skybreak handler groundWave

Grounded only. Cast has invulnerability from commit through last active tick; recovery is vulnerable. At activeStart query the forward volume, maximum 4 targets, excluding victims behind solid walls; R120 F4 U20. No actual terrain removal and no reference cinematic recreation. Heal caster by 60 HP once if any target takes damage. Only one hit per target. The entire form is spent on a miss. Visual rocks are client only.

Horizon Break handler longBeam

Grounded only. Same invulnerability window as Skybreak. Query beam against static cover and truncate at first solid obstacle. Targets with distance ≤8 m receive 100, otherwise 60, maximum 4; R90 F22 U5. No damage beyond obstruction or 32 m. Heal caster to full only if a near target takes damage; far hits heal 30 once. This finite damage rule deliberately replaces unclear source infinity and wind kill descriptions.

Orbit Collapse handler cinematicGrab

At active ticks select the nearest eligible victim, tie by entity ID. On capture both participants become protected from third parties. Anchor the sequence to the capture point; do not move collision capsules through the island. Schedule 100 damage 90 ticks after capture; release at +96 with R90 F6 U10, then recovery. On miss use the listed recovery with no capture extension. Heal caster to full on successful damage. A small cosmetic shockwave has no secondary damage. This simplified multi target treatment is an intentional local deviation.

All awakened attacks are noncounterable except the incoming hit that triggers Stillpoint Reversal. Setting consumesForm does not cancel the cast in progress. Damage caps remain 100 HP per target; do not encode JavaScript Infinity.

15 Stone Vanguard activation and context

Awakening activation sv.activate Ground Resonance

Required meter=200 damage points; gain 1 point per actual HP dealt in Base. This explicitly resolves conflicting meter multipliers locally. Idle, grounded and unprotected are required. Activation lasts 30 ticks with armor, no invulnerability, no damage and no heal. On completion set form=Awakened, meter=0, reset awakened use bits, contextSpent=false. There is no time expiry. Death, accepted kit change, or use of a slotted awakening consumes it. Natural health regeneration remains off.

P1 contextual attack selection

While awakened and contextSpent=false, the following inputs replace one M1 action. Priority is held urn → wall opportunity → grounded mini uppercut finisher. They do not occupy skill slots. contextSpent becomes true at commit even on a miss. An accepted contextual move never consumes the slotted awakening opportunity.

ID and label	Trigger	Timing and local outcome
sv.x1 Echo Tap	Fourth M1 with jumpHeld, grounded, no urn or wall target	Startup 12, active 4, recovery 30; 24 HP; box 2 by 2 by 2 m; U; R84 F14 U6
sv.x2 Gale Rebuff	Fourth M1 with valid wall target	Startup 12, capture up to 60, recovery 24; 40 HP at capture+48; U; R90 F18 U3
sv.x3 Meteor Cast	Attack while holding an urn	Startup 10, projectile life 120, recovery 24; 15 HP; radius 0.4; speed 28 m/s; U; R60 F10 U3

All three share once per form contextSpent; there is no additional cooldown. Echo Tap and Meteor Cast are counterable; Gale Rebuff is counterable at capture, not after capture. No invulnerability except capture protection for Gale Rebuff. Launch flags are local outcomes, not literal source choreography. On projectile miss the urn still returns to its anchor after 600 ticks.

Urn interaction shared by all kits

When not attacking, standing within 1.2 m of an unheld urn and pressing Attack with joystick magnitude <0.1 picks it up after 12 ticks instead of M1. Show a small “Pick up” label on Attack only while eligible. Holding an urn makes the next Attack throw it: ordinary throw damage 8, startup 10, speed 18, sphere radius 0.3, B, R36 F6 U2. Guard drops it harmlessly; damage, capture or death drops it. Ground pickup is not available in P0.

Passive and finisher policy

Faster awakening is represented by the 200 point requirement; other kits require 350. No unknown extra stat boost is inferred. Cosmetic knockout variations use the lethal hit’s existing tick and damage. No gore, bodily dismemberment, copied impact frames or recognizable anime sequence. The counter status is private to the owner; the local UI shows its remaining duration, other players see only the original stance animation.

15 Tide Duelist base abilities

ID and label	Startup active recovery	Cooldown	Damage HP	Range and flags
td.b1 Current Bind	10 5 18	1020	24 total	2.4 m U; capture
td.b2 Undertow Spiral	14 5 24	1200	19 total	2.2 m B; capture
td.b3 Tidal Cast	12 4 18	900	9 plus collision 3	2.2 m U; capture
td.b4 Mirror Current	0 60 30	1020	22 response	Counter; U response

Current Bind handler captureBarrage

On initial capture schedule damage [4,4,4,4,8] at relative ticks [0,10,20,30,42]. Give caster armor on capture; victim is susceptible to third party ordinary damage. Release at +43 with R60 F7 U3. Supports airborne ragdoll but not grounded ragdoll. If victim dies mid sequence stop remaining damage and release. Capture duration extends the initial active window only after a successful hit; miss transitions directly to recovery. Acceptance: a missed initial probe cannot attach a later victim during recovery.

Undertow Spiral handler dragSlam

On capture, caster armor; schedule 5 damage at +0 and 14 at +48. Move the pair forward by 3 m over 36 ticks using swept collision; hitting a wall stops motion but not the remaining schedule. Release at +49 with R72 F3 U8. Supports airborne and grounded ragdoll. P1 elevated branch: if cast started at least 2 m above the surface and lands before +48, increase final damage to 24, once. This local cap replaces unbounded slope velocity exploits. Acceptance: crossing a wall never drags the victim through it.

Tidal Cast handler throwCapture

Capture supports all three target states; apply 9 damage at +18, then release with R60 F10 U2 at +19. Attach one collision bonus of 3 for the next 90 ticks if the victim hits a static wall, never the floor. The thrown victim can hit one other living player for 5, B, stun 18; that third party damage cannot trigger a counter or another thrown body collision. The first wall or player collision also deals the 3 bonus to the thrown victim, then clears the collision marker; never apply it twice. Caster armor only during capture. Acceptance: two nearby walls cannot multiply the 3 damage bonus.

Mirror Current handler counter

Open window [commit,commit+60). On a counterable melee hit, cancel the incoming hit, capture its attacker within 4 m with line of sight, then deal [6,16] at trigger+[12,30]; release at +31 with R60 F8 U3. Caster has armor, not invulnerability. If melee attacker is beyond 4 m, absorb and end stance without response damage. A reflectable projectile triggers the shared reflection rule, ends stance and grants no automatic melee response. Response is noncounterable. On whiff, recovery is fully vulnerable. No held button prolongs the window.

15 Tide Duelist awakening and variants

Awakening td.activate Ocean Surge

Required meter 350; 60 tick grounded activation; armor during startup, invulnerability ticks 48–59. At tick 48 deal 20 HP in radius 5 m, U, armorBreak, noncounterable, R72 F6 U6, max 4. Heal 40 HP once at completion. Awakened duration is 2940 ticks from completion. Timers continue during all abilities and disconnection. A committed cast completes after form expiry, but new awakened casts cannot start. Base cooldown clocks continue running in parallel.

ID and label	Startup active recovery	Cooldown	Damage HP	Range and flags
td.a1 Cascade Chain	8 5 16	510	26 total	2.5 m U; capture
td.a2 Deepfall	16 5 30	Once per form	69	2.5 m U; cinematic; armorBreak
td.a3 Split Tide	16 18 24	840	38 or 26 miss branch	3 m U
td.a4 Riptide Pursuit	12 30 30	570	27 ranged or 30 close	Travel 9 m U; capture

Cascade Chain captures all target states; damage [4,4,4,4,10] at capture+[0,8,16,24,32]. Caster gains armor on capture. Release +33, R60 F5 U4. No area damage. Deepfall gives startup armor, then protects captured participants. Damage at capture+72; release +78, R90 F4 U12. On final impact other targets within 4 m take 20, U, noncounterable, R60 F6 U4 once. Maximum total targets 4; original victim cannot also take area damage. A miss spends the one use.

Split Tide grants startup armor through active end. Probe at offsets 0 and 17; first hit deals 12, stun 30, second 26, R72 F10 U4. Same victim can receive both for 38. If first misses, the second remains valid for 26. No capture; counterable; no armor break. A first hit interrupted by armorBreak prevents the second.

Riptide Pursuit selects close branch if an eligible target is within 2 m at commit. Otherwise dash forward at 18 m/s for up to 30 active ticks, stop at wall or capture. Ranged capture deals 27 at +36; close capture deals [8,22] at +0,+36. Release +37, R72 F8 U5. Caster armor only after capture. P1 close branch may continue once toward a second distinct target within 6 m after release: 12 tick sweep, capture then 31 damage at +30. Never recapture the same life; total cast duration cap 150 ticks; no extra cooldown reset. Counterable initial captures; no copied character specific clash.

Finisher passive heals caster once per victim life: b1 35, b2 30, b3 20, b4 40, a1/a2/a3 50, a4 zero. Only a lethal hit to a living enemy qualifies; corpse farming is deliberately excluded.

15 Ember Artificer base abilities

ID and label	Startup active recovery	Cooldown	Damage HP	Range and flags
ea.b1 Cinder Array	10 40 24	900	21 standing or 13 ragdoll	2.5 m B; max 3
ea.b2 Vent Rush	10 24 24	1020	15 front; 10 rear	Travel 8 m; radius 2.5 m B
ea.b3 Ember Orb	Charge 0–180 then 6 release	1500	15 or 25 full	Projectile 40 m, radius 0.35
ea.b4 Furnace Drop	18 48 30	1020	20	Impact radius 4 m B

Cinder Array uses hit offsets [0,8,16,24,32,39], standing damage [2,2,2,2,3,10], ragdoll damage [1,1,1,1,2,7]. Snapshot target branch at its first hit, not on every tick. Final hit R72 F3 U9. moveScale=0.2, turnRate=90 degrees/second. In air hold caster vertical velocity at zero through active ticks. Armor none.

Vent Rush moves along commit aim at 20 m/s during active ticks, with sweep and wall stop. A second press after 6 active ticks ends travel early; pressing before that is ignored. Rear blast at activeStart: radius 2 m centred 1.5 m behind, 10 HP, U, R36 F4 U2. Front blast at travel end: sphere radius 2.5 centred 1.5 m ahead, 15 HP standing or 24 ragdoll, chip 6, R72 F10 U3. A target may take only the larger of the two phases: reserve rear candidates and apply rear damage only if not hit by front, at travel end. No gain from duplicated phases. Air aim pitch is clamped −45 to +45 degrees; floor collisions end motion.

Ember Orb is holdRelease. Server starts charge on accepted down, releases on up or after 180 ticks; every hold packet refreshes the lease. A stale input lease cancels charge with no shot and keeps cooldown. Partial charge damage 15, B with chip 5; only full 180 tick charge deals 25, U. moveScale=0.15, turnRate=120 degrees/second, vertical velocity frozen in air. On release: 6 tick windup, spawn at capsule front clear of walls, speed 24 m/s, life 120 ticks, path has a fixed 0.5 m lateral arc easing back to centre at 20 m. Sphere sweep each tick. On collision radius 2 m splash, max 4, damage applied once; R60 F6 U3. Reflectable; direct and splash do not stack.

Furnace Drop rises with velocity +10 for 18 ticks, then aims toward a point up to 8 m ahead and descends at 18 m/s. Impact on first ground contact within 48 active ticks; otherwise expire into normal fall without explosion. Damage 20, chip 4, R84 F4 U10. Interruption cancels damage and restores gravity. Aerial start follows the same cap.

15 Ember Artificer awakening abilities

Awakening ea.activate Stellar Furnace

Meter requirement 350; grounded activation 90 ticks. Armor throughout; no invulnerability. Radius 5 m damage pulses at 0 and 72 ticks, each 16 HP, U, noncounterable, R60 F4 U3; maximum 4 targets and one hit per pulse. Heal 40 at completion. Duration 2820 ticks from completion. Caster cannot gain meter from activation. This chooses the reported 47 second duration as a D tuning value, not verified parity.

ID and label	Startup active recovery	Cooldown	Damage HP	Geometry and flags
ea.a1 Arc Descent	30 18 30	900	40 hit or 20 miss	Swept reach 8 m; U
ea.a2 Comet Lock	20 24 36	1200	50 standing or 30 ragdoll	Sweep 16 m; U; noncounterable
ea.a3 Furnace Ray	30 90 30	1500	35 maximum	Beam length 24, radius 1 m; B
ea.a4 Solar Vent	72 180 48	Once per form	70 near or 30 far maximum	Beam length 36, radius 3 m; U

Arc Descent advances at 24 m/s during active ticks, stopping on wall. Startup armor; initial hit counterable. On capture grant cinematic protection; deal 15 immediately and 25 at +36, release +42 R84 F5 U8. If initial probe misses, enter 30 tick vulnerable blast windup then explode radius 5 for 20, U, noncounterable, R60 F5 U5. If interrupted during that windup, emit one smaller radius 2 blast for 13 before cancellation; if killed, emit none. No miss branch if a captured victim dies after capture.

Comet Lock travels at 40 m/s for up to 24 ticks. Startup vulnerable; dash and capture invulnerable. On hit capture all target states; branch is fixed at capture. Deal branch damage at +54; nearby other targets within 4 m take 20, chip 5, R72 F5 U6. Victim excluded from splash. Release +60 with R90 F3 U8. Motion and capture are bounded by static collision. On miss enter vulnerable recovery. Noncounterable does not mean it can hit invulnerable targets.

Furnace Ray emits seven pulses at [0,15,30,45,60,75,89], each 5 HP, max 4 targets; no more than 35 per target. Clip beam at solid cover; turnRate=30 degrees/second; movement zero, vertical velocity zero. Gain armor after first confirmed damage through active end. Each pulse stuns 18 and pushes 1 m/s along the beam; chip 1 each. Counterable; interruption clears pending pulses.

Solar Vent emits ten pulses at [0,20,40,60,80,100,120,140,160,179]. At each pulse distance ≤10 m deals 7, otherwise 3; total cap 70 per target. U, counterable, no armor; turnRate=15 degrees/second; vertical velocity zero. Stun 24, forward velocity 2 m/s. Cover truncates the beam. Miss or interruption spends the use. No cumulative wall trapping damage beyond the ten pulses.

16 Shared variants passives and handler tests

Cyborg passive equivalents

Extended fourth M1: Ember Artificer uses depth 2.6 m and launch F10 U4; mini uppercut U12. Other hits retain universal values. Grounded evasive creates one radius 2.5 m blast on evasive tick 0 for 14 HP, B, noncounterable, stun 18, max 4; airborne evasive has no blast. The local counterability choice resolves the source conflict explicitly.

On a lethal ability hit, reset ea.b1–b4 and ea.a1–a2 cooldown end ticks to the current tick. Never reset ea.a3 or ea.a4, once use bits or evasive. One reward per victim life, awarded to the actual killer only. Deferred hit phases from the same cast cannot trigger a second reset. A room restart does not restore dead bodies as rewardable targets.

P1 wall attack shared handler

With the section 13 wall condition, replace the fourth M1 with wallCombo. Startup 8, capture window 3, scheduled hits [3,3,6] at capture+[0,12,30], release +36, recovery 18, cooldown 360 ticks. Initial capture B and counterable; subsequent hits cannot be countered. Caster armor during capture. Release victim R60 F7 U3 away from wall. For Vanguard awakened context, Gale Rebuff takes priority and uses its own values. Kit specific original animations share this deterministic behavior; exact reference wall choreography and damage are not reproduced.

Handler implementation interface

```
interface AbilityHandler {
  canStart(ctx: CombatContext, def: AbilityDef): Reject | null;
  start(ctx: CombatContext, def: AbilityDef): Cast;
  step(ctx: CombatContext, cast: Cast): HitIntent[];
  recast?(ctx: CombatContext, cast: Cast): void;
  release?(ctx: CombatContext, cast: Cast): void;
  cancel(ctx: CombatContext, cast: Cast, why: CancelReason): void;
}
```

Cast stores immutable initial aim, tick, generation and definition hash; mutable phase, hit registry, captured life ID, branch, deadlines and local counters. cancel must be idempotent and restore gravity, input permission, capture protection and temporary armor. Never change a shared AbilityDef at runtime. Use a fresh cast instance.

Mandatory per ability tests

Generate tests for every 24 slotted and 3 context IDs: accepted use; cooldown rejection; correct phase ticks; miss; valid hit; no duplicate damage; correct block behavior; counter behavior; interrupted startup; caster death; victim death; static occlusion; ragdoll eligibility; form expiry; two clients observing the same result. Charge and recast add cancellation, lease timeout and duplicate edge tests. Cinematic captures add owner removal and simultaneous capture tests. Beam tests hold a target against a wall for the full duration and assert the damage cap.

Reference quirks intentionally excluded: corpse farming, infinite damage values, client owned hit claims, wall tunnelling, physics exploit damage amplification and copied cinematic clashes. Maintain these as documented deviations, not unresolved implementation choices.

16 Ability normalization and handler boundaries

This page supplies implementation defaults for cases where a card describes a sequence rather than one fixed hit window. It is part of the local contract and takes precedence over generic shape defaults for these handlers.

Cast timing and data completion

A cast starts at commit tick c; active interval is [c+startup,c+startup+active). A probe that captures extends the cast through captureTick+lastScheduledOffset+1, then recovery begins. On miss, recovery begins at the end of the probe window. Cooldown is measured from c even when the cast has a charge or capture extension. For once per form actions use cooldown=0 plus oncePerForm=true; it does not mean reusable. A consuming Vanguard move has both oncePerForm and consumesForm. All other slotted skills have consumesForm=false.

Ember Orb has variable charge stage 0–180 ticks, then startup=6, active=1 spawn tick, recovery=24. The projectile remains alive independently of recovery. Meteor Cast has startup=10, active=1, recovery=24; its projectile life is 120. A dead or disconnected owner does not erase an already fired projectile; it retains attribution to that owner life and expires normally. It cannot grant meter, healing or cooldown reset to a new life. This overrides removal of owner effects on death only for detached projectiles; held beams and scheduled capture damage always cancel.

Exact special shapes

Dash/capture sweeps use sphere radius 0.6 m at capsule centre+0.2 m Y, swept over the previous to next authoritative centre. This applies to Arc Descent, Comet Lock and Riptide Pursuit. Their stated range limits total caster travel from activeStart; stop at the smaller of that range, active duration travel or solid collision. All spherical area attacks centre on the stated impact or caster position, use 3D distance and static line of sight. No target is hit across an island slab.

Skybreak’s box is width 20, height 8, depth 24, centred 12 m forward and 2 m above caster centre. Other box attacks use section 14. Beams begin 0.6 m forward of capsule centre+0.4 m Y and extend along allowed aim. Clip the beam against static geometry before testing player capsules. Arc Descent’s missed blast and any collateral area effect exclude the caster. Generic area events target at most four opponents; melee barrage maxTargets is per cast, not per pulse.

Remaining rules

All cooldowns persist while in another form; base slots display their own timers on return. New awakening resets only awakened use bits and contextSpent, not base cooldowns. Activation is accepted only from grounded Idle and does not cleanse stun. Meter gain is floor(actualDamage100/100) points with fractional remainder retained in meter100; no artificial rounding loss per pulse. Player meter100 thresholds are 20000 or 35000, while HUD percent is threshold normalized.

Temporary gravity suspension sets vY=0 each tick while active and restores ordinary gravity on exit; it does not restore the old velocity. Prop pickups, wall actions and M1 finishers share the offensive permission check. Counterable defaults remain true unless an explicit card says otherwise; armor is not damage reduction. P1 generic finishers are cosmetic on lethal damage only, so they do not require unverified original HP thresholds.

17 Architecture and dependency baseline

Process boundaries

The iPad runs a Vite built TypeScript client with Three.js rendering and a local Rapier world for movement prediction against static colliders. The Mac Studio runs one Node.js process with Colyseus rooms, authoritative Rapier worlds and an HTTP session service. One same origin HTTPS entry serves the client, HTTP endpoints and WebSocket upgrade. No Redis, database server or multi process matchmaking is required for four players.

The authoritative world is headless. It never imports Three.js, browser APIs or asset loaders. Shared modules contain serializable types, pure arithmetic, data validators and collision descriptions. Client presentation consumes events and snapshots; it cannot mutate server state. Data registries are bundled and hashed. Both sides refuse a connection with incompatible protocolVersion or contentHash.

Version baseline observed on 1 November 2026

These versions were read from the npm registry during specification research. Availability does not prove mutual compatibility; task T01 must compile and run the adapter smoke test before the lockfile is accepted. Use exact versions, no caret or tilde. [T09]

Package	Version	Role
colyseus	0.18.10	Rooms, authenticated matchmaking, transport
@colyseus/sdk	0.18.9	Browser client and reconnection
@colyseus/schema	5.0.43	Replicated state encoding
@dimforge/rapier3d-compat	0.21.0	Shared static collision and server queries
three	0.187.0	WebGL2 renderer
typescript	7.0.2	Type checks and build
vite	8.3.5	Client build
vitest	5.0.5	Unit and headless integration tests
@playwright/test	1.64.1	Browser journeys and layout checks
zod	4.6.7	Strict runtime message validation

Use Node 24 LTS, pnpm workspace and a committed lockfile; pin the exact Node 24 patch and pnpm version in T01 after checking their official release metadata. Do not silently downgrade dependencies or combine an old Colyseus tutorial with 0.18 imports. Verify @types/three, transport and test adapter versions against package peer dependencies; record exact resolutions in docs/dependencies.md.

Colyseus adapter decision

Use 0.18 room lifecycle and reconnection hooks. This specification uses a custom fixed step accumulator inside setTimestep(callback, 1000/60) and custom input packets. Do not also enable built in prediction or a second physics scheduler. patchRate=50 ms, maxClients=4, maxMessagesPerSecond=90. onDrop reserves 20 seconds; onReconnect restores the same life; onLeave cleans up. API signatures are checked in T01 because docs and published packages can drift. [T01 T02 T03]

18 Project tree and module ownership

```
elemental-battlegrounds/
  AGENTS.md
  package.json  pnpm-workspace.yaml  pnpm-lock.yaml
  tsconfig.base.json  .node-version  .env.example
  apps/
    client/
      index.html  vite.config.ts
      src/
        main.ts  app/AppFlow.ts  app/SettingsStore.ts
        input/PointerRouter.ts  input/IntentSampler.ts
        input/ControlLayout.ts  input/KeyboardRouter.ts
        net/RoomClient.ts  net/ClockSync.ts
        net/Prediction.ts  net/SnapshotBuffer.ts
        scene/GameScene.ts  scene/CameraRig.ts
        scene/FighterView.ts  scene/EffectsPool.ts
        ui/Hud.ts  ui/Lobby.ts  ui/Settings.ts
        assets/AssetManifest.ts  audio/AudioBus.ts
    server/
      src/
        main.ts  http/sessionRoutes.ts
        auth/InviteStore.ts  auth/SessionStore.ts
        rooms/ArenaRoom.ts  rooms/PracticeRoom.ts
        state/ArenaSchema.ts  state/PlayerSchema.ts
        sim/World.ts  sim/FixedClock.ts
        sim/Movement.ts  sim/ActionMachine.ts
        combat/HitResolver.ts  combat/CaptureRegistry.ts
        combat/MeterSystem.ts  combat/RespawnSystem.ts
        abilities/AbilityRunner.ts  abilities/handlers/
        net/ValidateInput.ts  net/RateLimits.ts
        ops/Health.ts  ops/StructuredLog.ts
  packages/shared/src/
    protocol.ts  schema.ts  math.ts  movement.ts
    config/combat.ts  config/abilities.ts
    config/arena.ts  config/abilityOverrides.ts
    evidence/referenceCatalog.ts
  assets/
    fighters/  animations/  effects/  audio/  maps/
    manifest.json  provenance.csv
  tests/
    unit/  combat/  integration/  browser/
    fixtures/  load/  device/checklist.md
  ops/
    launchd/com.elemental.arena.plist
    scripts/start.sh  scripts/stop.sh  scripts/smoke.sh
    scripts/invite.ts  README.md
  docs/
    spec.md  decisions/  dependencies.md
    reference-observations.md  control-calibration.json
    tasks.md  test-results/  asset-guidelines.md
```

Ability handlers are named functions, not a switch that grows without tests. Keep handler state in Cast, world authority in sim, render objects in scene, and DOM listeners in input/ui. Every long lived object has dispose(). Package boundaries are enforced through import rules. No client imports apps/server or secret bearing modules. shared/evidence is build excluded from the browser bundle so research names are not accidentally shipped as UI labels.

19 Network message contracts

Protocol version is 1. Transport is Colyseus over secure WebSocket. Packets are reliable and ordered by transport, but validate duplicates and reconnect epochs at the application boundary. Never trust a client claimed player ID; derive identity from the authenticated Client.

```
type Edge = { id:number; kind:"jump"|"dash"|"skillDown"
  |"skillUp"|"awaken"; slot?:1|2|3|4 };
interface InputPacket {
  v:1; epoch:number; seq:number; sampleTick:number;
  moveX:number; moveZ:number; yaw:number; pitch:number;
  attackHeld:boolean; blockHeld:boolean; jumpHeld:boolean;
  runHeld:boolean; chargeHeldSlot:0|1|2|3|4;
  edges:Edge[];
}
interface ActionResult {
  epoch:number; seq:number; edgeId:number;
  accepted:boolean; tick:number; castId?:string;
  reason?:"STATE"|"COOLDOWN"|"METER"|"FORM"|"INVALID";
}
interface CombatEvent {
  eventId:string; tick:number; kind:"cast"|"hit"|"block"
    |"counter"|"capture"|"release"|"ko"|"respawn";
  actorId:string; targetId?:string; lifeId:number;
  moveId?:string; damage100?:number; position?:Vec3;
}
```

Send input at 30 Hz; send a new packet immediately for edges if the rate bucket permits. Max 45 input packets/second sustained, burst 15. Max 6 edges per packet, 20 per second sustained, burst 8. moveX/Z and yaw/pitch must be finite; clamp vector length to 1, pitch to the permitted ability range, and angular delta to the applicable turn rate. seq and edge IDs are monotonic uint32 within an epoch; renew the epoch before wrap. Reject unknown keys, wrong version, stale epoch and packets >1024 bytes.

Input coordinates are world horizontal vectors already transformed from camera space; the server still normalizes them. sampleTick is diagnostic only and cannot schedule a past or future hit. Process latest held values at the next server tick, and unique edges in arrival order under section 12 priorities. An old seq cannot overwrite newer holds. Ack the newest processed seq in owner state; send one ActionResult per edge. M1 hold repeats have server generated cast IDs, no edge ID.

Other client messages: ready {kitId,contentHash}, changeKit {kitId}, ping {nonce,clientMs}, leave {}. Server messages: welcome {epoch,serverTick,tickRate,contentHash}, actionResult, combatEvent, pong {nonce,serverMs,serverTick}, shutdown {deadlineMs,reason}. Rate limits: ready/changeKit 2 per second; ping 1 per second; no arbitrary client events. All message schemas are versioned and tested independently of UI.

20 Replicated state and HTTP interface

State schema

RoomState contains protocolVersion, contentHash, serverTick, mapId, phase, players and projectiles. Player records contain entityId, lifeId, kitId, position xyz, velocity xyz, facingYaw, hp100, action, actionStartTick, actionEndTick, moveId, form, formEndTick, meter100, cooldownEndTicks[4], awakenedUsedMask, contextSpent, connected, protectedUntilTick, kills, deaths and assists. Owner only fields include ackSeq, ackInputTick, epoch, pendingKit, evasiveEndTick and chargeStartTick. Do not replicate invites, tokens or internal auth identifiers.

Projectiles contain entityId, castId, ownerId, ownerLifeId, spawnTick, position, velocity, chargeClass, remainingTicks and reflectionCount. They are authoritative gameplay entities; visual debris is not replicated. On each 20 Hz patch include the current serverTick alongside positional fields. Encode positions to centimetres and yaw to 1/1024 turn in schema adapter code; the simulation retains full precision. Health is integer hundredths; cooldowns are absolute ticks, not decremented network counters.

HTTP endpoints

Route	Authentication and request	Response and errors
GET /healthz	None; no details	200 {ok:true}; 503 if not ready
GET /api/version	None	200 {protocolVersion,contentHash,buildId}
POST /api/session	Same origin; {invite:string,displayName:string}; max 1 KB	201 with HttpOnly session cookie and {expiresAt}; 401 invalid, 429 rate limited
POST /api/join-ticket	Valid session cookie; {roomCode:string}; CSRF header	200 {ticket,expiresAt,roomName}; 401/403/409/429
DELETE /api/session	Cookie plus CSRF	204; revoke session and disconnect sockets
GET /api/csrf	Valid cookie; same origin	200 {csrfToken}; no cache

A join ticket is random 256 bit, one use, valid 30 seconds, bound to session, room, content hash and protocol. Pass it as a join option; Colyseus onAuth consumes it atomically. The room cannot be joined merely by guessing its ID or calling joinOrCreate directly. Allow only configured room names; server owns room creation. Cookie is Secure, HttpOnly, SameSite=Strict, Path=/, 8 hour idle session with 24 hour absolute expiry. Refresh idle expiry only on authenticated activity.

Display names are plain text 1–16 Unicode grapheme clusters; reject control characters and render with textContent. Presets Arel and Nora are conveniences, not identities. Invitation tokens never appear in logs. Invitation links use a URL fragment consumed by the Welcome page and removed with history.replaceState before requests. No token in query strings, analytics or referrers.

Session and invitation hashes are stored in a small local SQLite file or an atomic JSON store for P0; choose atomic JSON as baseline. Write temp file, fsync and rename; permissions 0600. Server memory remains the authority for active rooms; no combat persistence is required.

21 Fixed simulation loop and movement

Use a monotonic clock, dt=1/60 second, at most 5 catch up steps per scheduler callback. Track overload separately. Physics and combat each run exactly once per logical tick; render rate cannot accelerate cooldowns. Rapier world.timestep is dt. The official character controller computes corrected displacement against geometry; gameplay supplies gravity, jump and knockback. [T04 T05]

```
onScheduler(elapsedMs):
  accumulator += min(elapsedMs, 250)
  steps = 0
  while accumulator >= 1000/60 and steps < 5:
    tick += 1
    expireStatesAndReservations(tick)
    drainValidatedIntents(tick)
    clearHoldsIfLeaseExpired(tick, 15)
    runActionTransitions(tick)
    for player in stableEntityOrder:
      desired = integrateMovementAndAbilityMotion(player, dt)
      corrected = controller.computeMovement(desired, staticWorld)
      stageKinematicPosition(player, corrected)
    physics.step()
    hits = querySweptAttacksAndProjectiles(tick)
    resolveHitsInStableOrder(hits)
    releaseInvalidCaptures(); processDeathsAndRespawns()
    publishSchemaFields(); recordReplayChecksum()
    accumulator -= 1000/60; steps += 1
  if accumulator >= 1000/60:
    recordOverload(); accumulator = min(accumulator,1000/60)
```

Dropping excess accumulated time slows simulation relative to wall time under overload; it never grants extra hits or accelerates timers. Alert if this occurs more than 3 times in 10 seconds. Stop accepting new players while overloaded. Match timers are simulation ticks; invite/session/reconnect deadlines use monotonic wall time. No request may submit its own dt.

Motion integration

Player collider is position based kinematic. vHorizontal approaches desiredVelocity by acceleration*dt, unless action motion overrides it. vY += gravity*dt unless grounded or a handler suspends gravity. desiredTranslation = velocity*dt; character controller adjusts it against static geometry. Apply next kinematic translation, then step. Filter player capsules out of movement collision to avoid client prediction dependence on remote poses; combat still hits their hurt capsules. Local players can overlap but cannot push each other through cover.

Knockback replaces velocity components using the hit origin’s horizontal direction; if zero distance, use attacker forward. Ragdoll is an authoritative capsule with ballistic motion, not a networked skeleton. Visual limbs are procedural only. Ground normal dot up must exceed cos(45°) to count grounded. Detect landing from previousGrounded=false to currentGrounded=true with downward prior velocity.

Use seeded random only for cosmetic spawn choices that are not covered by a deterministic rule. Replay records validated inputs, seed, config hash and server build; never promise cross platform bitwise Rapier determinism. Test numerical tolerance and state invariants on the pinned runtime.

21 Hit resolver and protocol adapter pseudocode

These are algorithms to implement, not complete game source. Keep all shape queries and state mutation on the server. The network adapter queues intent; it never calls damage directly.

```
validateAndQueue(client, packet):
  player = authenticatedPlayerFor(client)
  require strictSchema(packet) and boundedPacketSize(packet)
  require packet.epoch == player.inputEpoch
  require packet.seq > player.lastReceivedSeq
  consumePacketRateTokenOrReject(client)
  require finiteAndBoundedVectors(packet)
  for edge in packet.edges:
    if edge.id > player.lastEdgeId:
      consumeActionRateTokenOrReject(client)
      queue(edge, arrivalTick, player.entityId)
      player.lastEdgeId = edge.id
  replaceHeldIntent(player, normalized(packet))
  player.lastReceivedSeq = packet.seq
  player.lastInputTick = tick

resolve(hit):
  if invalidLifeOrCastGeneration(hit): return
  if hitAlreadyRecorded(hit.cast, hit.target, hit.phase): return
  if deadOrProtectedOrOutOfRangeOrOccluded(hit): return
  if not eligibleTargetState(hit): return
  recordHitOnce(hit)
  if eligibleCounter(hit):
    consumeIncomingHit(); triggerCounterOrReflect(); return
  if blockableAndInGuardCone(hit):
    applyChipAndBlockstun(); emitBlock(); return
  damage = min(hit.damage100, target.hp100)
  target.hp100 -= damage
  creditDamageToCurrentLife(hit.owner, damage)
  if not armorStopsReaction(hit): applyStunOrRagdoll(hit)
  if target.hp100 == 0: finalizeKOOnce(target.lifeId)
  emitHit(hit, damage)

cancelCast(cast):
  if cast.cancelled: return
  cast.cancelled = true; cast.generation += 1
  cancelFutureHitPhases(cast)
  releaseOwnedCaptureIfAny(cast)
  removeTemporaryArmorAndGravityOverride(cast)
  enterRecoveryUnlessDead(cast.owner)
```

A reflected detached projectile retains its new owner and that owner’s life ID. Meter and passive rewards require that same owner life to remain alive. The schema/state adapter publishes after all tick mutations. Do not emit a mutable object reference into asynchronous logging or event delivery; serialize an immutable copy. Event IDs are roomEpoch:serverTick:counter. Client event dedup keeps a bounded 2048 entry ring. State is authoritative even when a cosmetic event was missed; a late join reconstructs current actions from moveId and actionStartTick rather than replaying old explosions.

Transport room hooks use onCreate for registry/world initialization, onAuth for ticket consumption, onJoin for entity allocation, onDrop/onReconnect for seat preservation, onLeave for permanent removal and onDispose for world cleanup. No authentication secrets belong in replicated schema. Validate actual installed Colyseus signatures in T01, with a real join, patch, drop and reconnect smoke test.

22 Prediction interpolation and reconnection

Local prediction

Predict locomotion against the shared static collider map at 60 Hz using the same movement function. Keep up to 120 tick samples of movement intent and predicted pose. Do not predict damage, capture, KO, meter, passive rewards or successful counters. A provisional cast animation may start, but the server acceptance tick controls its real phase.

A 30 Hz input sample represents held intent until superseded. Owner snapshots also include acknowledged input receive tick so replay can anchor the held interval. For replay, store the client's estimated server tick and duration of that intent, not simply one movement step per packet. Server ackSeq confirms intent processing, not that the two machines stepped at identical times. On owner snapshot, reset the prediction base to authoritative position/velocity/action at snapshot.serverTick and replay locally buffered held inputs for subsequent estimated ticks, capped at 12 ticks. Discard edges already acknowledged; never replay damage side effects.

If positional error ≤0.1 m, ease visual offset over 6 frames; between 0.1 and 2 m, correct simulation immediately and ease visual mesh over 100 ms. Above 2 m or on capture, teleport, death or respawn, snap and clear old prediction. Collision logic always uses corrected pose, never smoothed mesh pose. When stunned or captured, suspend locomotion prediction until a newer server action permits it.

Remote presentation and clocks

Buffer snapshots for 100 ms, interpolate position and shortest path yaw. Extrapolate at most 100 ms with last velocity, then freeze and show connection uncertainty. Never extrapolate damage. Estimate server time from a ping midpoint, median of the last 8 samples; clamp clock corrections to 5 ms/frame unless reconnecting. Display cooldown using estimated serverTick but let server rejection settle discrepancies.

Connection lifecycle

On unexpected loss show "Reconnecting" and disable input. Clear all held states and provisional actions. Server onDrop sets connected=false, neutralizes input and reserves the seat for 20 wall clock seconds. Avatar remains present, vulnerable and subject to void/death rules; no disconnect escape. After 20 seconds onLeave removes it and releases captures. Graceful leave has no reservation. [T03]

On successful reconnect receive a full authoritative state, increment input epoch, reset seq/edge IDs, clear snapshot and prediction buffers, and resume only after the new welcome arrives. Preserve health, cooldowns, form and scores; a dead avatar stays dead until the usual respawn deadline. Disable SDK offline gameplay message buffering or drop queued old epoch packets explicitly. Keep a reconnection token in sessionStorage for reload recovery, never localStorage; redact it from logs.

If the server restarts, room state is lost. Show "Arena restarted" and return to the lobby using the still valid session if available; no silent reconstruction of stale combat. Content changes require reload after leaving the room, never hot swap collision or abilities mid session.

23 Rendering animation assets and audio

Scene and lifecycle

One WebGLRenderer, one scene, one camera per active game. Use WebGL2; initialize Rapier before joining. Render independently with requestAnimationFrame. Resize canvas from its displayed CSS rectangle and update camera aspect; cap the internal pixel count rather than blindly using devicePixelRatio. Dispose materials, textures, geometries, audio nodes, listeners and room subscriptions on exit. [T07 T08]

Asset contract

Each fighter has an independently authored glTF 2.0 model with one consistent rig and named clips: idle, walk, run, jump, fall, land, guard, dashFront, dashSide, dashBack, hitLight, ragdollPose, getup, death, plus the ability IDs. Use one shared rig when practical. Gameplay timing is stored in ability data; clip duration is retimed to the server phase, with markers used only for visual and sound effects.

Required manifest entry: id, path, SHA256, byteSize, license, author, sourceURL or commission reference, creationMethod, rigVersion and bounds. Empty provenance fails the production build. Keep original source files for art and audio. No extraction from Roblox, wiki GIFs, anime footage, game sounds or community asset rips. Reference names exist only in research records. A renamed asset alone does not establish independent creation.

Art limits and readable combat

P0 uses original geometric fighters and simple effects with the final scale and silhouettes; P1 replaces them without changing hitboxes. Use cel shading with three tonal bands, one directional light and an ambient hemisphere. Maximum one shadow casting light, shadow map 1024 on medium and disabled on low. Avoid full screen post processing on low; no motion blur. Telegraph large attacks with ground shape and charging silhouette visible before damage.

For each move create an original windup, active and recovery pose. Big ground attacks use stone fans, circular fractures and original motion rhythm. Water moves use ribbon arcs. Heat moves use rings and heat particles. No source camera shot recreation, logos, sound samples or title typography. KO effects are dust, light and a falling silhouette with no gore.

Audio and effects

Unlock AudioContext on Play. Sound event is keyed by combatEvent.eventId; deduplicate across reconnect and replay. Maximum 16 simultaneous one shots; prioritize local hits, then nearby major abilities, then ambience. Spatial attenuation reaches zero by 45 m. Limit combined loudness through a master compressor; maintain volume sliders. No autoplay music before interaction.

All effects use pools and an expiry tick. Maximum 200 active particles on low, 600 medium, 1000 high; translucent layers should not cover the entire screen. Reduce particle density first under load. Gameplay telegraphs cannot be removed by quality scaling. Context loss pauses rendering and inputs, shows Recovering graphics, and rebuilds from authoritative state after restoration.

24 Performance budgets and hosting

Measurable budgets

Area	Target and required measurement
Two actual iPads	Median ≥55 FPS, p95 frame ≤25 ms during 10 minute ordinary play; no sustained <30 FPS for >3 s
Four player stress	p95 frame ≤33.3 ms for 5 minutes with overlapping awakenings on low quality
Client memory	JS heap steady state ≤150 MB where measurable; no upward trend >20 MB across 10 room joins
GPU budget	≤120 draw calls and ≤150k visible triangles on medium; 512 by 512 typical character textures, 1024 maximum
Rendering pixels	Low ≤0.75 million, medium ≤1.5 million, high ≤2.5 million; scale 0.6–1.5 relative to CSS
Initial load	Compressed shell ≤1 MB; playable P0 assets ≤15 MB; P1 full cache ≤35 MB
Server tick	Four players: p95 ≤4 ms, p99 ≤8 ms; no backlog sustained beyond 3 callbacks
Network	Steady per player upstream ≤8 KB/s, downstream ≤30 KB/s; burst ≤80 KB/s for 1 s
Interaction	Local button highlight <1 frame; LAN confirmed action p95 <100 ms; remote target p95 <200 ms at RTT≤120 ms

These are acceptance targets, not claims about unknown iPad models or Mac Studio specifications. Record hardware, browser and OS in the report. Safari may not expose reliable total memory; use available counters, repeated scene cycling and crash observations, and mark unavailable metrics explicitly.

Adaptive quality samples frame time over 120 frames. If p90 >28 ms for two windows, drop one quality level; if p90 <18 ms for 30 seconds, raise one. Minimum 15 seconds between changes. Never reduce simulation rate or hit detection accuracy. Asset LOD and particle density are client only.

Mac Studio topology

Default remote access is a private Tailscale network with the Mac Studio and both iPads enrolled. Tailscale Serve provides the HTTPS entry for the local game service; the iPads still play in Safari, but need the Tailscale app connected for remote access. Use a stable tailnet hostname and same origin HTTPS/WSS. Do not use Funnel for the private default. Tailscale documents Serve as private to the tailnet. [T10]

Node listens on 127.0.0.1:2567 and serves both static build and API/Colyseus transport. Proxy the entire origin, preserving Colyseus generated room WebSocket paths; do not assume only /ws requires upgrade. Keep the Mac on Ethernet, prevent sleep during sessions and ensure enough cooling. A Mac that sleeps cannot host an active game.

If the family requires access without a VPN app, implement a separately configured public HTTPS reverse proxy or authenticated tunnel with the same invitation system. This is an optional deployment profile, not part of the default setup. No router port forwarding is required for the private profile.

24 Deployment runbook and recovery

Build and start contract

Repository scripts MUST exist with these names: dev, build, typecheck, lint, test, test:integration, test:browser, test:load, content:validate, assets:audit and start:prod. A fresh install runs pnpm install --frozen-lockfile, pnpm build and pnpm start:prod without editing source code. Local development may expose a LAN port for testing only; production binds loopback behind HTTPS.

Environment keys: NODE_ENV=production, HOST=127.0.0.1, PORT=2567, PUBLIC_ORIGIN=https://configured-tailnet-host, ROOM_CAPACITY=4, DATA_DIR=an-owner-only-directory, LOG_LEVEL=info. Session secrets and invitation hashes live outside the repository. Validate all required configuration at startup and fail with a nonsecret explanation.

First deployment sequence

1. Record the Mac hardware, OS and runtime versions; build from a tagged commit and validate the content hash.

2. Start the production service on loopback. Check /healthz, /api/version, static assets, session exchange and a local two client room.

3. Configure the supported Tailscale Serve command for the installed version; the researched pattern is tailscale serve --bg http://127.0.0.1:2567. Verify with tailscale serve status. Restrict tailnet grants to the family devices and this HTTPS service. [T10]

4. Open the HTTPS URL on both iPads. Confirm certificate trust, WSS upgrade, audio activation and actual simultaneous touch behavior. Test once from another network, not only home Wi Fi.

5. Generate separate single use invitations with ops/scripts/invite.ts, validity 24 hours. Let the adult distribute them; do not send messages automatically. Run a full match and reconnect test.

6. Install the provided launchd job under the chosen service account with absolute runtime paths, WorkingDirectory, logs and restart on failure. Keep data and secrets restricted to that account.

Operations and updates

Log structured events: startup build hash, room counts, join/leave reason, tick percentiles, input rejects by category, overload and unexpected exceptions. No names, invitation tokens, session cookies, reconnection tokens or raw IPs in routine logs. Rotate daily, retain seven days, cap total logs at 100 MB. Health endpoint contains no player list. Metrics/admin UI is local only.

For upgrade: announce shutdown in game with 60 second notice, stop new joins, drain rooms, back up the small session store, switch the release directory and restart. Roll back by selecting the prior tagged build and compatible content/config; reject old clients until they reload. Do not migrate active combat state between builds.

On failure: show service unavailable, inspect health and redacted logs, confirm Mac awake and tailnet connection, then restart the service. A watchdog restart must not enter a crash loop; after five failures in five minutes stop and notify the operator. Verify shutdown releases all resources and exits within 10 seconds.

24 Security and privacy requirements

Authority and abuse resistance

The server accepts intentions only. Never expose RPCs for setPosition, damage, hitTarget, awardKill, refillMeter, resetCooldown, changePhysics or spawn arbitrary assets. Clamp movement and angular rate, verify state transitions, check every cooldown on the server and reject expired or repeated action IDs. Private networking reduces exposure but does not replace game validation.

Boundary	Required control	Acceptance probe
Session exchange	Hashed single use 256 bit invitations; 5 attempts/min per source plus 30/min global	Reuse and brute force fail without a valid cookie
HTTP and WebSocket	Exact Origin allowlist; TLS; max bodies; tickets bound to room/session	Foreign origin and missing ticket rejected
Messages	Strict schemas, finite numbers, length limits, per action rate buckets	NaN, huge arrays, unknown keys and invalid slots cannot crash room
Assets	Same origin content hashed files; no user supplied URLs	A client cannot request an arbitrary remote asset through server
Admin	Local operator scripts, no browser admin API in P0/P1	A player token cannot create invites or run room commands
Secrets	Outside source and client bundle; redacted logs	Build artifact and log scans find no tokens or secret values

Use a restrictive Content Security Policy: default-src self; script-src self plus only the WebAssembly allowance required by the tested Safari build; style-src self; img-src self blob data as needed; connect-src self wss to the configured origin; object-src none; frame-ancestors none; base-uri none. Avoid unsafe-eval and inline scripts. If the installed runtime needs wasm-unsafe-eval, document and test that exact exception. Security headers include nosniff and a no-referrer policy. CSRF applies to cookie authenticated mutation routes; Origin validation also applies to WebSocket upgrades.

Privacy and child friendly scope

Store only a pseudonymous session ID, preset/custom display label, local preferences and in memory session scores. No real names required, no email address, birthday, location, ads, analytics SDK, public profiles or chat. Arel and Nora can use their preferred labels without exposing them to a public service. Delete expired session records daily and allow an operator to revoke a session instantly.

Availability limits

One room defaults to four players; at most four rooms in P1 if load tests pass. Max 32 projectiles per room and max 4 active casts per player including persistent statuses. Expensive area queries cap candidates at room population; cosmetic debris never generates server entities. Bound every collection and timer. Disconnect a client after 10 invalid messages in 10 seconds; record the category, not payload. Rate limiting must tolerate legitimate five finger input and reconnection bursts.

No automatic public publication, domain purchase or installation is performed by this specification. The deployment tasks describe the authorized implementation deliverable; external rollout remains an operator action.

25 Automated acceptance tests

Test IDs are stable. Use deterministic simulated ticks for combat, real sockets for protocol integration, and browser automation for UI. A passing unit suite alone is not evidence that touch controls work on an iPad.

ID	Fixture and action	Required result
A01	Two authenticated clients join same private room and select identical kits	Distinct entity and life IDs; both see each other and identical health
A02	Hold M1 for 180 ticks at a target in range	Four hit sequence follows legal timing; no attacks during fourth recovery; total per chain 14
A03	Fourth M1 grounded with Jump held; repeat airborne	First produces uppercut, second downslam; no simultaneous ordinary jump
A04	Guard at angles 59, 60 and 61 degrees from front	B hit blocks at ≤60; hits at 61; U always bypasses
A05	Request dash twice one tick apart; then left and right	Only first forward dash; side dashes share their bucket, independent of forward
A06	Evasive at ragdoll age 11 then 12 ticks	First rejected, second accepted if cooldown ready; landing bonus cleared
A07	Replay same skill edge 10 times, including stale epoch	One cast and one cooldown; stale epoch rejected
A08	Two attackers capture same target on same tick	Stable order chooses one; losing cast cannot damage as if it captured
A09	Fault Strike direct hit then evasive before landing	Exactly 20 damage, no later 10; without evasive exactly 30
A10	Ember Orb held 179 versus 180 ticks, release	Damage 15 versus 25; partial blockable, full bypass; one projectile
A11	Charge with missing input for 15 ticks	Cancelled without projectile; cooldown retained
A12	Beam target trapped at wall full duration	Furnace Ray ≤35; Solar Vent ≤70; no repeated extra hits
A13	Lethal hit plus another hit same tick	One KO, one passive reward, no negative HP
A14	Form expires while a capture is active	Committed cast finishes; no new awakened cast; all protections released
A15	Disconnect while captured; expire reservation	No stuck capture, no fresh HP on reconnect, seat released on expiry
A16	Malformed packet NaN, 100 edges, oversized string	Reject safely; room remains running; rate bucket enforces limits
A17	Cross origin join and ticket reused in another room	Authentication fails; no state or player list leaked
A18	Spawn anchors occupied or under active attack	Deterministic farthest valid selection; fallback protection works
A19	Throw urn through thin cover at maximum speed	Swept query hits cover; no damage behind cover
A20	Reconnect with queued old skillUp/skillDown	Old epoch discarded; no cast or stuck charge after resync

Every ability adds the matrix from section 16. T12, T13, T15, T16, T20 and T21 are split into subcommits .1 through .4, one per slotted move; T23 uses .1 through .3 for the contextual moves. Snapshot equality compares health, action, cooldowns, meter and ownership; rendering differences are irrelevant. Server tests must explicitly assert damage sums and time boundaries, not only “event emitted”. Keep tests independent from the implementation’s internal branch structure.

26 Device network and performance acceptance

Actual iPad checklist

Run on both Arel's and Nora's iPads in Safari, with installed iPadOS recorded. Also test a home screen shortcut if used. Browser emulation is useful but cannot certify physical multitouch, thermal behavior or Safari lifecycle handling.

ID	Manual scenario	Pass condition
D01	Landscape at 1024×768, 1180×820 and actual safe viewport	All buttons within safe rectangle, no overlap, touch targets meet minimum
D02	Move plus camera plus held M1 plus Jump	Inputs remain independent for 30 seconds; fourth hit variant works
D03	Charge skill while moving camera, release outside its button	One shot at server charge value; captured pointer does not activate another button
D04	Block, invoke app switcher, return	No stuck guard or movement; reconnect/neutral behavior visible
D05	Rotate, resize browser chrome, enter split view	Layout recalculates; no inherited pointer; small view shows rotate/enlarge panel
D06	Drag button layout, save, reload and reset	Stable normalized positions; invalid overlaps rejected; Reset restores fixture
D07	Start music and effects, background then return	Audio resumes only through supported user interaction; no overlapping loops
D08	Die during a cinematic; open Settings	Camera returns safely; no invisible UI or permanent invulnerability
D09	Ten minute match, then ten lobby/game cycles	Meets frame budget, no crash, no growing listener/entity counts
D10	Reference layout comparison in private Roblox session	Record deltas and settings; never mark exact parity without measurements

Network simulation matrix

Run A01, A07, A08, A12, A15 and A20 with RTT 20/80/150/250 ms, jitter 0/20/40 ms and 0/1/3 percent packet loss at the transport/network layer. WebSocket retransmission means loss often appears as delay; do not simulate unordered delivery as if the transport were UDP. At RTT≤150, confirm responsive prediction and no authority divergence. At 250 ms, allow delayed combat feedback but never duplicate damage, teleport cheating or stuck states.

Test Wi Fi to cellular hotspot switch, 2 second outage, 19 second outage, 21 second outage, browser reload and Mac service restart. Outages that recover within the reservation reconnect to the same life; beyond reservation returns to lobby. Network outage durations are measured from server drop detection; include detection time in the report.

Load and release evidence

Use four authenticated headless clients cycling melee, projectiles and awakenings for 15 minutes. Record server tick histograms, messages per second, bandwidth and heap trend. Separately run two physical iPads for 20 minutes to capture thermal effects. Save reports under docs/test-results with commit, content hash and device metadata. A release report lists each test PASS, FAIL or NOT RUN. Missing physical device tests remain NOT RUN; agents must never fabricate them.

27 Sequenced commits for the playable release

Each task should be one reviewable commit. If a task grows beyond roughly 500 changed nonasset lines, split it by module or handler while retaining its acceptance ID. Tests may be in the same commit as the behavior they establish. Do not defer all integration to the final phase.

Task and dependency	Files to create or change	Done means
T01 Bootstrap none	Root package files, docs/dependencies.md, minimal client/server entry	Exact lockfile; typecheck/build; one real Colyseus join using installed API
T02 Shared contracts T01	shared/protocol.ts, schema.ts, config/combat.ts	Strict validators and protocol compatibility tests
T03 Authentication T02	server/auth, http/sessionRoutes.ts, ops/scripts/invite.ts	Session exchange and single use join ticket; A17
T04 Room lifecycle T03	ArenaRoom, ArenaSchema, PlayerSchema, RoomClient	A01 and reconnect seat reservation with minimal state
T05 Arena colliders T02	config/arena.ts, map JSON, sim/World.ts	Collision hash match; bridge and spawn validator
T06 Movement T04 T05	shared/movement.ts, sim/Movement.ts, FixedClock	Fixed step capsule movement and jump; wall blocking
T07 Touch shell T02	PointerRouter, ControlLayout, IntentSampler, Hud	Layout fixture; independent pointer ownership; D01 simulated
T08 Camera and prediction T06 T07	CameraRig, Prediction, ClockSync, SnapshotBuffer	Reconciliation, camera collision and stale input lease tests
T09 Combat states T06	ActionMachine, HitResolver, CaptureRegistry	A04 A05 A06 A08; no orphan captures
T10 M1 and guard T07 T09	Universal handlers and combat tests	A02 A03; touch hold maps to legal server repeats
T11 Ability framework T09	AbilityRunner, ability schema, registry validator	Phase scheduler, hit deduplication, cancel idempotence
T12 Vanguard base T11	Four sv base handlers and definitions	A09 plus all four per move matrices
T13 Duelist base T11	Four td base handlers and definitions	Capture, throw and counter matrices
T14 Awakening system T12 T13	MeterSystem, form state, owner HUD	Damage based meter, expiry, death and use flags; A14
T15 Vanguard awakening T14	Four sv awakened handlers	One move consumes form, counter status and finite damage
T16 Duelist awakening T14	Four td awakened handlers	Captures, miss branch, one use and expiry tests
T17 Respawn and FFA T10 T14	RespawnSystem, scoreboard, lobby kit change	A13 A18; scoring, assist, spawn protection
T18 P0 playable gate T15 T16 T17	AppFlow, PracticeRoom, onboarding, tests	Two clients complete all moves; D02 D04 D08 on both iPads

T18 may use original geometric placeholder art. It cannot omit functionality, fake networking with local bots, or claim physical device tests that were not run. The first playtest is intentionally early enough to adjust control spacing before expensive animation work.

27 Sequenced commits for the complete scope

Task and dependency	Files to create or change	Done means
T19 Charge and projectile system T18	Projectile simulation, hold lease, projectile state	Sweeps, reflect cap, A10 A11 A19
T20 Artificer base T19	Four ea base handlers and definitions	Recast, air branches and charge release matrices
T21 Artificer awakening T20	Four ea awakened handlers and activation	A12; hit/miss/interrupted branches; form expiry
T22 Passives T21	Passive hooks, reward registry	One per life healing/reset; explicit exclusions
T23 Props and wall context T18 T19	Urn state, wall handler, sv context handlers	All three context moves; shared one use bit; pickup priority
T24 Duelist P1 variants T16	Elevated slam and assault chain	Two distinct targets maximum; no unbounded chain
T25 Original art pipeline T18	AssetManifest, provenance, rig and placeholder replacements	Asset audit and all required animation IDs
T26 Effects audio and camera T21 T23 T25	EffectsPool, AudioBus, original tracks	Effects caps, event dedup, reduced motion and D07 D08
T27 Settings and layout editor T07 T18	SettingsStore, layout editor, accessibility	D05 D06; validation and versioned persistence
T28 Production hosting T04 T18	ops scripts, launchd, same origin server	HTTPS/WSS from both iPads on remote network
T29 Security hardening T22 T28	RateLimits, origin checks, CSP, redaction	A16 A17; secrets scan and revocation
T30 Performance pass T24 T26 T27	LOD, pixel caps, pooling, load fixtures	Section 24 budgets measured; no combat rate changes
T31 Fidelity calibration T18 T27	reference observations, layout JSON, decision records	D10; discrepancies tagged C/X/U, defaults revised only deliberately
T32 Release candidate T29 T30 T31	Test reports, README, release manifest	All P0/P1 tests PASS or explicit approved limitations; tagged build

Task packet template

Task ID and objective; prerequisite commit hashes; allowed files; relevant specification section; exact behavior; tests to run; expected outputs; excluded scope. End with changed files, executed checks and remaining failures. Do not ask an agent to “finish combat” or “implement all abilities” without a list of IDs.

Definition of done for every commit

Build and type checks pass; task tests pass; no secret material; no invented source fact; no stale listeners or timer; no client authority leak; no unrelated refactor; task ledger updated. Asset only changes require provenance and visual review rather than meaningless unit tests. UI changes require a layout screenshot at the target viewport.

Change control

If a reference check changes a move observation, update the evidence ledger first. If local behavior must change, create a decision record containing old value, new value, reason, affected move IDs, protocol/content hash impact and tests. Rebalance only through shared configuration; do not scatter numeric overrides in animation code. A new dependency requires a brief compatibility check and exact version pin.

The implementing agent may resolve routine code structure without asking permission. It must not interpret a missing fidelity observation as permission to invent source mechanics, install public services or extend the roster.

28 Agent readable AGENTS instructions

Copy the following into the repository root AGENTS.md before T01. The document's own task is complete at delivery; these instructions govern later game implementation.

```
# Elemental Battlegrounds implementation contract

Read docs/spec.md and docs/tasks.md before changing code.
Implement one task ID at a time. Check its prerequisites first.
Treat Sonnet 5.5 Low and Sol 6.1 Low as selected execution agents.
Do not infer capabilities or change model settings from these names.

Authority order:
1. Explicit user requirements.
2. Specification local rules and numeric registry.
3. Per ability overrides.
4. Default handler semantics.
5. Acceptance tests consistent with the above.

Never claim an original game fact without an evidence ID.
D values are local design choices. C and X are not verified facts.
Do not copy Roblox or anime assets, audio, animation or UI artwork.
Use original shipped names. Keep reference labels in research only.
Do not expand the three fighter scope without a new specification.

Server decides position limits, hits, HP, cooldowns, forms and scores.
Clients send validated intent only. Never accept a target damage claim.
Use fixed 60 Hz simulation. Render timing never schedules damage.
All action timers use integer ticks. Auth expiry uses wall time.
Do not add setTimeout damage handlers or a second physics scheduler.
Every capture, listener, effect and room subscription has cleanup.
Use strict TypeScript and strict runtime message schemas.
Keep pure shared modules free of DOM, Three.js and server secrets.
Use exact dependency versions and the committed lockfile.

Before implementing an ability, add its definition and tests.
Test miss, block, counter, duplicate hit, interruption, death and expiry.
Test network behavior with real clients, not only mocked callbacks.
Physical iPad checks must be recorded as NOT RUN until actually run.
Do not report success from unexecuted tests or invented measurements.

Use rg for file/text discovery. Read relevant code before editing.
Keep each task small. No unrelated refactoring or package upgrades.
Resolve routine details autonomously within the approved contract.
If requirements conflict, document the conflict before choosing a rule.
For a broad new request, list concrete items and files and get scope
confirmation before exploring or editing beyond the current task.

Run pnpm typecheck, pnpm lint, relevant tests and pnpm build.
Run content:validate after data changes and assets:audit for assets.
Update docs/tasks.md with status and evidence, then commit coherently.
Report changed behavior, executed checks and actual limitations.
Never expose secrets in logs, chat, screenshots or version control.
Do not send invitations, deploy publicly or buy services automatically.

Keep each session to one task. When finished or around 50 steps,
recommend a fresh session with a 2 to 3 line handoff summary.
```

The task ledger uses TODO, IN PROGRESS, BLOCKED or DONE plus commit hash and test report path. BLOCKED must name a concrete missing dependency. A completed code task with pending physical checks is IMPLEMENTED, DEVICE CHECK PENDING rather than DONE.

29 Assumptions verification log and deviations

Assumptions to validate without blocking scaffolding

ID	Assumption or uncertainty	Current decision and closure evidence
V01	iPad models and OS unknown	Use WebGL2 baseline; record both physical devices at T18
V02	Home network and remote needs unknown	Private Tailscale default; confirm adult setup at T28
V03	Exact original touch rectangles unavailable	D layout in section 7; measure and compare at T31
V04	Default hold/Quick Use settings unclear	D Quick Use and auto run on; observe original settings
V05	Character pages have stale indexed content	Preserve C status; record build date for any live checks
V06	Hero meter multiplier differs	D requirement 200 versus 350; do not claim 2× or 3× parity
V07	Cyborg awakening duration differs	D 2820 ticks; compare source 42 and 47 seconds
V08	Evasive counterability differs	D noncounterable local blast; document as a deviation
V09	Detailed frame data not published	Every local tick and hitbox value is D, editable in data
V10	Current extra burst systems not corroborated	Exclude from P0/P1; create a P2 source task before adding
V11	Complete current roster uncertain	Three core kits only; other and restricted forms excluded
V12	Dependency API compatibility untested	T01 smoke test pins exact runtime and resolves peer versions

Explicit fidelity deviations

The game uses metres and finite damage, a small floating island arena, a four player default cap, original names and assets, no passive regeneration, simplified wall attacks, capped collateral damage, protected cinematic captures, deterministic hit arbitration, a limited prop system and no corpse farming. It omits reference map destruction, ranked modes, cosmetics economy, character specific cross kit cinematics and out of scope fighters. These choices are intentional and testable; none should be described as exact original mechanics.

What counts as verified later

A source check needs URL, publisher, access date, claim and quoted or paraphrased evidence. A measurement needs device, game setting, date, video frame rate, trial count and observed range. Record disagreements without deleting old evidence. A guide repeating a wiki is not independent corroboration. Never infer a missing number from an animation or a marketing phrase.

Release gates

P0: T01–T18 complete, two physical iPads connect and fight, both kits fully usable, no authority or stuck input bugs. P1: T19–T32 complete, third kit and context attacks present, original assets audited, remote play and performance tested. Fidelity label: only use “closely matched controls” after T31; never use “exact replica” based on this document alone.

Completion means a reproducible build plus the required test evidence, not merely code generation. The source and assumption ledgers remain living project records after implementation.

30 Game research sources

All sources accessed or searched on 1 November 2026. Fandom citations may refer to searchable indexed text when direct retrieval failed; their crawl age is not a live build verification. Short factual summaries are used here; no reference artwork or audio is included.

[R01] Yielding Arts. The Strongest Battlegrounds official Roblox experience. Primary creator listing. Establishes game identity, mobile support and core PC action mapping; not touch dimensions or numeric combat data.

https://www.roblox.com/games/10449761463/The-Strongest-Battlegrounds

[R02] The Strongest Battlegrounds Controls PC Mobile and Console. Community guide, dated 11 August 2026. Supports touch action vocabulary; not an authoritative layout measurement.

https://thestrongestbattleground.wiki/tools/controls/

[R03] The Strongest Hero. The strongest battlegrounds community wiki. Indexed character move set, Serious Mode behavior and older balance values. Direct fetch failed during research; indexed text used.

https://the-strongest-battlegrounds.fandom.com/wiki/The_Strongest_Hero

[R04] The Strongest Hero. The Strongest Battlegrounds Rblx community wiki. Additional contextual attacks, variants and differing numbers. Indexed text used; current live parity unverified.

https://the-strongest-battlegrounds-rblx.fandom.com/wiki/The_Strongest_Hero

[R05] Hero Hunter. The Strongest Battlegrounds Rblx community wiki. Core kit, awakening, passive and explicit unfinished/rework/archive warnings. Indexed text used.

https://the-strongest-battlegrounds-rblx.fandom.com/wiki/Hero_Hunter

[R06] Hero Hunter Garou Moves Damage and Ultimate. RBLXGUIDE. Derivative community guide; useful for variant inventory, not independent confirmation of live balance.

https://rblxguide.com/games/the-strongest-battlegrounds/characters/hero-hunter

[R07] Destructive Cyborg. The strongest battlegrounds community wiki. Older ability, charge, passive and duration descriptions; indexed text used.

https://the-strongest-battlegrounds.fandom.com/wiki/Destructive_Cyborg

[R08] Destructive Cyborg. The Strongest Battlegrounds Rblx community wiki. Updated variant and passive descriptions, including conflicting awakening duration and evasive details; indexed text used.

https://the-strongest-battlegrounds-rblx.fandom.com/wiki/Destructive_Cyborg

[R09] All TSB Characters Moves Damage and Ultimates. RBLXGUIDE. Index corroborates the three contextual Hero attacks, but its roster counts include differing categories.

https://rblxguide.com/games/the-strongest-battlegrounds/characters

[R10] Updates. The Strongest Battlegrounds Rblx community wiki. Historical reports of auto run, held auto use skills and additional Hero context moves; not proof of present defaults.

https://the-strongest-battlegrounds-rblx.fandom.com/wiki/Updates

30 Additional sources and engineering references

[R11] The Strongest Hero. Strongest Battlegrounds Roblox community wiki. Reports a conflicting three times awakening passive; used only to document uncertainty.

https://strongest-battlegrounds-roblox.fandom.com/wiki/The_Strongest_Hero

[R12] Basic Combat. The Strongest Battlegrounds Rblx community wiki. Reports held M1 strings and fourth hit variation vocabulary. Character specific exceptions are not adopted without separate verification.

https://the-strongest-battlegrounds-rblx.fandom.com/wiki/Basic_Combat

[T01] Colyseus. Rooms documentation, version 0.18. Primary source for room loop and patch configuration. Local simulation and packet design are decisions of this specification.

https://docs.colyseus.io/room

[T02] Colyseus. Migrating to 0.18. Documents setTimestep replacing the older name and built in prediction alternatives; this design uses one custom scheduler.

https://docs.colyseus.io/migrating/0.18

[T03] Colyseus. Reconnection. Primary lifecycle documentation for onDrop, allowReconnection, onReconnect and final leave behavior.

https://docs.colyseus.io/room/reconnection

[T04] Dimforge. Rapier JavaScript Controllers, version 0.21 guide. Character controller movement correction and kinematic usage.

https://rapier.rs/docs/user_guides/javascript/character_controller/

[T05] Dimforge. Rapier JavaScript Scene queries. Primary documentation for ray and shape queries. Local hit semantics are not Rapier defaults.

https://rapier.rs/docs/user_guides/javascript/scene_queries/

[T06] W3C. Pointer Events. Primary standard for pointer ownership, capture and touch action behavior.

https://www.w3.org/TR/pointerevents/latest/

[T07] Three.js. Responsive Design. Canvas CSS sizing, camera aspect and drawing buffer limits.

https://threejs.org/manual/pages/responsive.html

[T08] Three.js. WebGLRenderer documentation. Rendering and resource lifecycle reference.

https://threejs.org/docs/pages/WebGLRenderer.html

[T09] npm package registry. Package publisher metadata retrieved directly for the ten package versions listed in section 17. Query pattern is registry.npmjs.org followed by package name and /latest; record the exact lockfile at implementation time.

https://registry.npmjs.org/colyseus/latest

https://registry.npmjs.org/@colyseus/sdk/latest

https://registry.npmjs.org/@dimforge/rapier3d-compat/latest

https://registry.npmjs.org/three/latest

[T10] Tailscale. tailscale serve command. Primary reference for private tailnet service exposure and HTTPS proxy configuration. Verify installed CLI behavior during deployment.

https://tailscale.com/docs/reference/tailscale-cli/serve